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PROSTHETIC VALVE DOCKING DEVICE

Abstract

Example docking devices for securing a prosthetic valve at a native valve are disclosed. In certain examples, a docking device can include a coil having a plurality of helical turns when deployed at the native valve, and a retention member covering at least a portion of the coil. The retention member can include a first axial segment. In certain examples, the first axial segment encloses only a partial circumference of the coil. In certain examples, the first axial segment includes a first circumferential portion and a second circumferential portion, and the second circumferential portion has a smoother outer surface than the first circumferential portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of PCT Patent Application No. PCT/US2023/082578, filed Dec. 5, 2023, which claims the benefit of U.S. Provisional Application No. 63/387,911, filed Dec. 16, 2022, each of which is incorporated by reference in its entirety herein.

FIELD
[0002] The present disclosure concerns examples of a docking device configured to secure a prosthetic valve at a native heart valve, as well as methods of assembling such devices.
BACKGROUND
[0003] Prosthetic valves can be used to treat cardiac valvular disorders. Native heart valves (for example, the aortic, pulmonary, tricuspid and mitral valves) function to prevent backward flow or regurgitation, while allowing forward flow. These heart valves can be rendered less effective by congenital, inflammatory, infectious conditions, etc. Such conditions can eventually lead to serious cardiovascular compromise or death. For many years, the doctors attempted to treat such disorders with surgical repair or replacement of the valve during open heart surgery.
[0004] A transcatheter technique for introducing and implanting a prosthetic heart valve using a catheter in a manner that is less invasive than open heart surgery can reduce complications associated with open heart surgery. In this technique, a prosthetic valve can be mounted in a compressed state on the end portion of a catheter and advanced through a blood vessel of the patient until the valve reaches the implantation site. The valve at the catheter tip can then be expanded to its functional size at the site of the defective native valve, such as by inflating a balloon on which the valve is mounted or, for example, the valve can have a resilient, self-expanding frame that expands the valve to its functional size when it is advanced from a delivery sheath at the distal end of the catheter. Optionally, the valve can have a balloon-expandable, self-expanding, mechanically expandable frame, and/or a frame expandable in multiple or a combination of ways.
[0005] In some instances, a transcatheter heart valve (THV) may be appropriately sized to be placed inside a particular native valve (for example, a native aortic valve). As such, the THV may not be suitable for implantation at another native valve (for example, a native mitral valve) and/or in a patient with a larger native valve. Additionally, or alternatively, the native tissue at the implantation site may not provide sufficient structure for the THV to be secured in place relative to the native tissue. Accordingly, improvements to THVs and the associated transcatheter delivery apparatus are desirable.
SUMMARY
[0006] The present disclosure relates to methods and devices for treating valvular regurgitation and/or other valve issues. Specifically, the present disclosure is directed to a docking device configured to receive a prosthetic valve and the methods of assembling the docking device and implanting the docking device.
[0007] A docking device for securing a prosthetic valve at a native valve can include a coil comprising a plurality of helical turns when deployed at the native valve. In addition to these features, a docking device can further comprise one or more of the components disclosed herein.

[0008] In certain examples, a docking device can include a retention member covering at least a portion of the coil.
[0009] In certain examples, the retention member includes a first axial segment which encloses only a partial circumference of the coil.
[0010] In certain examples, the retention member includes a first axial segment which includes a first circumferential portion and a second circumferential portion. The second circumferential portion has a smoother outer surface than the first circumferential portion.
[0011] Certain aspects of the disclosure concern an implant assembly including a radially expandable and compressible prosthetic valve and the docking device described above.
[0012] A method for assembling a docking device configured to receive a prosthetic valve can include preparing a retention member having a first axial segment and a second axial segment. The method can further include attaching the retention member to a coil of the docking device so that the first axial segment encloses only a partial circumference of the coil and the second axial segment encloses a full circumference of the coil. The coil is movable from a substantially straight configuration to a helical configuration.
[0013] A method for assembling a docking device configured to receive a prosthetic valve can include preparing a retention member having a tubular configuration. The retention member has a first axial segment and a second axial segment. The method can further include attaching the retention member to a coil of the docking device so that the retention member encloses a full circumference of the coil. The coil is movable from a substantially straight configuration to a helical configuration. The first axial segment has a first circumferential portion and a second circumferential portion. The second circumferential portion has a smoother outer surface than the first circumferential portion.
[0014] Certain aspects of the disclosure concern a method for implanting a prosthetic valve. The method can include deploying the docking device described above at a native valve, and deploying the prosthetic valve within the docking device.
[0015] The above method can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (for example, with body parts, heart, tissue, etc. being simulated).
[0016] In some examples, a docking device comprises one or more of the components recited in Examples 1-26 described in the section “Additional Examples of the Disclosed Technology” below.
[0017] The foregoing and other objects, features, and advantages of the disclosed technology will become more apparent from the following detailed description, which proceeds with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0018] FIG. **1** schematically illustrates a first stage in an exemplary mitral valve replacement procedure where a guide catheter and a guidewire are inserted into a vasculature of a patient and navigated through the vasculature and into a heart of the patient, towards a native mitral valve of the heart.
[0019] FIG. **2**A schematically illustrates a second stage in the exemplary mitral valve replacement procedure where a docking device delivery apparatus extending through the guide catheter is used to deploy a docking device at the native mitral valve.
[0020] FIG. **2**B schematically illustrates a third stage in the exemplary mitral valve replacement procedure where the docking device of FIG. **2**A is fully implanted at the native mitral valve of the patient and the docking device delivery apparatus has been removed from the patient.
[0021] FIG. **3**A schematically illustrates a fourth stage in the exemplary mitral valve replacement

procedure where a prosthetic heart valve delivery apparatus extending through the guide catheter is deploy a prosthetic heart valve within the implanted docking device at the native mitral valve.
[0022] FIG. **3**B schematically illustrates a fifth stage in the exemplary mitral valve replacement procedure where the prosthetic heart valve is fully implanted within the docking device at the native mitral valve and the prosthetic heart valve delivery apparatus has been removed from the patient.
[0023] FIG. **4** schematically illustrates a sixth stage in the exemplary mitral valve replacement procedure where the guide catheter and the guidewire have been removed from the patient.
[0024] FIG. **5**A is a side perspective view of a docking device in a helical configuration, according to one example.
[0025] FIG. **5**B is a top view of the docking device depicted in FIG. **5**A.
[0026] FIG. **5**C is a cross-sectional view of the docking device taken along line **5**C-**5**C depicted in FIG. **5**B, according to one example.
[0027] FIG. **5**D is a cross-sectional view of the docking device taken along the same line as in FIG. **5**C, except in FIG. **5**D, the docking device is in a substantially straight delivery configuration.
[0028] FIG. **6**A is a perspective view a prosthetic valve, according to one example.
[0029] FIG. **6**B is a perspective view of the prosthetic valve of FIG. **6**A with an outer cover, according to one example.
[0030] FIG. **7**A is a perspective view of an exemplary prosthetic implant assembly comprising the docking device depicted in FIG. **5**A and the prosthetic valve of FIG. **6**B retained within the docking device.
[0031] FIG. **7**B is a side elevation view of the prosthetic implant assembly of FIG. **7**A
[0032] FIG. **8** depicts a section of an example docking device in a substantially straight delivery configuration (the guard member is removed).
[0033] FIG. **8**A depicts a cross-section view of the docking device of FIG. **8** taken along the line **8**A-**8**A, according to one example.
[0034] FIG. **8**B depicts a cross-section view of the docking device of FIG. **8** taken along the line **8**B-**8**B, according to one example.
[0035] FIG. **9** depicts a section of another example docking device in a substantially straight delivery configuration (the guard member is removed).
[0036] FIG. **9**A depicts a cross-section view of the docking device of FIG. **9** taken along the line **9**A-**9**A, according to one example.
[0037] FIG. **9**B depicts a cross-section view of the docking device of FIG. **9** taken along the line **9**B-**9**B, according to one example.
[0038] FIG. **10** is a side view of an example docking device in a substantially straight delivery configuration, where a retention member of the docking device has a circumferential gap.
[0039] FIG. **11** is a side view of an example docking device in a helical deployed configuration, where a retention member of the docking device has a smooth circumferential portion facing radially outwardly and a textured circumferential portion facing radially inwardly.
[0040] FIG. **12**A depicts a holder device, according to one example.
[0041] FIG. **12**B depicts a retention member retained in the holder device.
[0042] FIG. **12**C depicts applying heat to smooth out an exposed portion of the retention member of FIG. **12**B, according to one example.
[0043] FIG. **12**D depicts the retention member of FIG. **12**B after the heat treatment.
DETAILED DESCRIPTION
General Considerations
[0044] It should be understood that the disclosed examples can be adapted to deliver and implant prosthetic devices in any of the native annuluses of the heart (for example, the pulmonary, mitral, and tricuspid annuluses), and can be used with any of various delivery approaches (for example, retrograde, antegrade, transseptal, transventricular, transatrial, etc.).

[0045] For purposes of this description, certain aspects, advantages, and novel features of the examples of this disclosure are described herein. The disclosed methods, apparatus, and systems should not be construed as being limiting in any way. Instead, the present disclosure is directed toward all novel and nonobvious features and aspects of the various disclosed examples, alone and in various combinations and sub-combinations with one another. The methods, apparatus, and systems are not limited to any specific aspect or feature or combination thereof, nor do the disclosed examples require that any one or more specific advantages be present or problems be solved. The technologies from any example can be combined with the technologies described in any one or more of the other examples. In view of the many possible examples to which the principles of the disclosed technology may be applied, it should be recognized that the illustrated examples are only preferred examples and should not be taken as limiting the scope of the disclosed technology.

[0046] Although the operations of some of the disclosed examples are described in a particular, sequential order for convenient presentation, it should be understood that this manner of description encompasses rearrangement, unless a particular ordering is required by specific language set forth below. For example, operations described sequentially may in some cases be rearranged or performed concurrently. Moreover, for the sake of simplicity, the attached figures may not show the various ways in which the disclosed methods can be used in conjunction with other methods. Additionally, the description sometimes uses terms like “provide” or “achieve” to describe the disclosed methods. These terms are high-level abstractions of the actual operations that are performed. The actual operations that correspond to these terms may vary depending on the particular implementation and are readily discernible by one of ordinary skill in the art.

[0047] As used in this application and in the claims, the singular forms “a,” “an,” and “the” include the plural forms unless the context clearly dictates otherwise. Additionally, the term “includes” means “comprises.” Further, the terms “coupled” and “connected” generally mean electrically, electromagnetically, and/or physically (for example, mechanically or chemically) coupled or linked and does not exclude the presence of intermediate elements between the coupled or associated items absent specific contrary language.

[0048] As used herein, the term “proximal” refers to a position, direction, or portion of a device that is closer to the user and further away from the implantation site. As used herein, the term “distal” refers to a position, direction, or portion of a device that is further away from the user and closer to the implantation site. Thus, for example, proximal motion of a device is motion of the device away from the implantation site and toward the user (for example, out of the patient's body), while distal motion of the device is motion of the device away from the user and toward the implantation site (for example, into the patient's body). The terms “longitudinal” and “axial” refer to an axis extending in the proximal and distal directions, unless otherwise expressly defined.

[0049] Directions and other relative references (for example, inner, outer, upper, lower, etc.) may be used to facilitate discussion of the drawings and principles herein, but are not intended to be limiting. For example, certain terms may be used such as “inside,” “outside,”, “top,” “down,” “interior,” “exterior,” and the like. Such terms are used, where applicable, to provide some clarity of description when dealing with relative relationships, particularly with respect to the illustrated examples. Such terms are not, however, intended to imply absolute relationships, positions, and/or orientations. For example, with respect to an object, an “upper” part can become a “lower” part simply by turning the object over. Nevertheless, it is still the same part and the object remains the same. As used herein, “and/or” means “and” or “or,” as well as “and” and “or.”

Exemplary Transcatheter Heart Valve Replacement Procedure

[0050] Described herein are various systems, apparatuses, methods, or the like, that can be used in or with delivery apparatuses to deliver a prosthetic implant (for example, a prosthetic valve, a docking device, etc.) into a patient body.

[0051] In certain examples, a delivery apparatus can be configured to deliver and implant a docking device at an implantation site, such as a native valve annulus. The docking device can be configured to more securely hold an expandable prosthetic valve implanted within the docking device, at the native valve annulus. For example, a docking device can provide or form a more circular and/or stable anchoring site, landing zone, or implantation zone at the implant site, in which a prosthetic valve can be expanded or otherwise implanted. By providing such anchoring or docking devices, replacement prosthetic valves can be more securely implanted and held at various valve annuluses, including at the mitral annulus which does not have a naturally circular cross-section.

[0052] In some examples, the docking device can be arranged within an outer shaft of the delivery apparatus. A sleeve shaft can cover or surround the docking device within the delivery apparatus and during delivery to a target implantation site. A pusher shaft can be disposed within the outer shaft, proximal to the docking device, and configured to push the docking device out of the outer shaft to position the docking device at the target implantation site. The sleeve shaft can also surround the pusher shaft within the outer shaft of the delivery apparatus. After positioning the docking device at the target implantation site, the sleeve shaft can be removed from the docking device and retracted back into the outer shaft of the delivery apparatus.

[0053] Fluid (for example, a flush fluid, such as heparinized saline or the like) can be provided to a pusher shaft lumen defined within an interior of the pusher shaft, a delivery shaft lumen defined between the sleeve shaft and the outer shaft of the delivery apparatus, and a sleeve shaft lumen defined between the pusher shaft and the sleeve shaft. By providing a consistent flow of fluid through these lumens of the delivery apparatus, stagnation of blood within the delivery apparatus can be reduced or avoided, thereby reducing a risk of thrombus formation.

[0054] An exemplary transcatheter heart valve replacement procedure which utilizes a first delivery apparatus to deliver a docking device to a native valve annulus and then a second delivery apparatus to deliver a prosthetic heart valve (for example, THV) inside the docking device is depicted in the schematic illustrations of FIGS. **1-4**.

[0055] As introduced above, defective native heart valves may be replaced with THVs. However, in certain instances, such THVs may not be able to sufficiently secure themselves to the native tissue (for example, to the leaflets and/or annulus of the native heart valve) and may undesirably shift around relative to the native tissue, leading to paravalvular leakage, valve malfunction, and/or other issues. Thus, a docking device may be implanted first at the native valve annulus and then the THV can be implanted within the docking device to help anchor the THV to the native tissue and provide a seal between the native tissue and the THV.

[0056] FIGS. **1-4** depict an exemplary transcatheter heart valve replacement procedure (for example, a mitral valve replacement procedure) which utilizes a docking device **52** and a prosthetic heart valve **62**, according to one example. During the procedure, a user can create a pathway to a patient's native heart valve using a guide catheter **30** (FIG. **1**). The user can deliver and implant the docking device **52** at the patient's native heart valve using a docking device delivery apparatus **50** (FIG. **2**A) and then removes the docking device delivery apparatus **50** from the patient **10** after implanting the docking device **52** (FIG. **2**B). The user can then implant the prosthetic heart valve **62** within the implanted docking device **52** using a prosthetic valve delivery apparatus **60** (FIG. **3**A). Thereafter, the user can remove the prosthetic valve delivery apparatus **60** from the patient **10** (FIG. **3**B), as well as the guide catheter **30** (FIG. **4**).

[0057] FIG. **1** depicts a first stage in a mitral valve replacement procedure, according to one example. As shown, the guide catheter **30** and a guidewire **40** can be inserted into a vasculature **12** of a patient **10** and navigated through the vasculature **12**, into a heart **14** of the patient **10**, and toward the native mitral valve **16**. Together, the guide catheter **30** and the guidewire **40** can provide a path for the docking device delivery apparatus **50** and the prosthetic valve delivery apparatus **60** to be navigated through and along, to the implantation site (for example, the native mitral valve **16** or native mitral valve annulus).

[0058] Initially, the user may first make an incision in the patient's body to access the vasculature **12**. For example, as illustrated in FIG. **1**, the user may make an incision in the patient's groin to access a femoral vein. Thus, in such examples, the vasculature **12** may include a femoral vein.
[0059] After making the incision to access the vasculature **12**, the user may insert the guide catheter **30**, the guidewire **40**, and/or additional devices (such as an introducer device or transseptal puncture device) through the incision and into the vasculature **12**. The guide catheter **30** (which can also be referred to as an "introducer device," "introducer," or "guide sheath") can be configured to facilitate the percutaneous introduction of various implant delivery devices (for example, the docking device delivery apparatus **50** and the prosthetic valve delivery apparatus **60**) into and through the vasculature **12** and may extend through the vasculature **12** and into the heart **14** but may stop short of the native mitral valve **16**. The guide catheter **30** can comprise a handle **32** and a shaft **34** extending distally from the handle **32**. The shaft **34** can extend through the vasculature **12** and into the heart **14** while the handle **32** can remain outside the body of the patient **10** and can be operated by the user in order to manipulate the shaft **34** (FIG. **1**).
[0060] The guidewire **40** can be configured to guide the delivery apparatuses (for example, the guide catheter **30**, the docking device delivery apparatus **50**, the prosthetic valve delivery apparatus **60**, additional catheters, or the like) and their associated devices (for example, docking device, prosthetic heart valve, and the like) to the implantation site within the heart **14**, and thus may extend all the way through the vasculature **12** and into a left atrium **18** of the heart **14** (and in some examples, through the native mitral valve **16** and into a left ventricle of the heart **14**) (FIG. **1**).
[0061] In some instances, a transseptal puncture device or catheter can be used to initially access the left atrium **18**, prior to inserting the guidewire **40** and the guide catheter **30**. For example, after making the incision to access the vasculature **12**, the user may insert a transseptal puncture device through the incision and into the vasculature **12**. The user may guide the transseptal puncture device through the vasculature **12** and into the heart **14** (for example, through the femoral vein and into the right atrium **20**). The user can then make a small incision in an atrial septum **22** of the heart **14** to allow access to the left atrium **18** from the right atrium **20**. The user can then insert and advance the guidewire **40** through the transseptal puncture device within the vasculature **12** and through the incision in the atrial septum **22** into the left atrium **18**. Once the guidewire **40** is positioned within the left atrium **18** and/or the left ventricle **26**, the transseptal puncture device can be removed from the patient **10**. The user can then insert the guide catheter **30** into the vasculature **12** and advance the guide catheter **30** into the left atrium **18** over the guidewire **40** (FIG. **1**).
[0062] In some instances, an introducer device can be inserted through a lumen of the guide catheter **30** prior to inserting the guide catheter **30** into the vasculature **12**. In some instances, the introducer device can include a tapered end that extends out a distal tip of the guide catheter **30** and that is configured to guide the guide catheter **30** into the left atrium **18** over the guidewire **40**. Additionally, in some instances the introducer device can include a proximal end portion that extends out a proximal end of the guide catheter **30**. Once the guide catheter **30** reaches the left atrium **18**, the user can remove the introducer device from inside the guide catheter **30** and the patient **10**. Thus, only the guide catheter **30** and the guidewire **40** remain inside the patient **10**. The guide catheter **30** is then in position to receive an implant delivery apparatus and help guide it to the left atrium **18**, as described further below.
[0063] FIG. **2**A depicts a second stage in the exemplary mitral valve replacement procedure where a docking device **52** can be implanted at the native mitral valve **16** of the heart **14** of the patient **10** using a docking device delivery apparatus **50** (which may also be referred to as an "implant catheter," or a "docking device delivery device," or simply "delivery apparatus").
[0064] In general, the docking device delivery apparatus **50** can include a delivery shaft **54** (which may also be referred to as an "outer shaft"), a handle **56**, and a pusher assembly **58**. The delivery shaft **54** can be configured to be advanced through the patient's vasculature **12** and to the implantation site (for example, native mitral valve **16**) by the user, and may be configured to retain

the docking device **52** in a distal end portion **53** of the delivery shaft **54**. In some examples, the distal end portion **53** of the delivery shaft **54** can retain the docking device **52** therein in a substantially straightened delivery configuration.

[0065] The handle **56** of the docking device delivery apparatus **50** can be configured to be gripped and/or otherwise held by the user to advance the delivery shaft **54** through the patient's vasculature **12**. Specifically, the handle **56** can be coupled to a proximal end of the delivery shaft **54** and can be configured to remain accessible to the user (for example, outside the body of the patient **10**) during the docking device implantation procedure. In this way, the user can advance the delivery shaft **54** through the patient's vasculature **12** by exerting a force on (for example, pushing) the handle **56**. In some examples, the delivery shaft **54** can be configured to carry the pusher assembly **58** and/or the docking device **52** with it as it advances through the patient's vasculature **12**. In this way, the docking device **52** and/or the pusher assembly **58** can advance through the patient's vasculature **12** in lockstep with the delivery shaft **54** as the user grips the handle **56** and pushes the delivery shaft **54** deeper into the patient's vasculature **12**.

[0066] In some examples, the handle **56** can comprise one or more articulation members **57** that are configured to aid in navigating the delivery shaft **54** through the vasculature **12**. For example, the one or more articulation members **57** can comprise one or more of knobs, buttons, wheels, and/or other types of physically adjustable control members that are configured to be adjusted by the user to flex, bend, twist, turn, and/or otherwise articulate a distal end portion **53** of the delivery shaft **54** to aid in navigating the delivery shaft **54** through the vasculature **12** and/or within the heart **14**.

[0067] The pusher assembly **58** can be configured to deploy and/or implant the docking device **52** at the implantation site (for example, the native mitral valve **16**). For example, the pusher assembly **58** can be configured to be adjusted by the user to push the docking device **52** out of the distal end portion **53** of the delivery shaft **54**. A pusher shaft of the pusher assembly **58** can extend through the delivery shaft **54** and can be disposed adjacent to the docking device **52** within the delivery shaft **54**. In some examples, the docking device **52** can be releasably coupled to the pusher shaft of the pusher assembly **58** via a connection mechanism of the docking device delivery apparatus **50** such that the docking device **52** can be released after being deployed at the native mitral valve **16**. Because the docking device **52** is retained by, held, and/or otherwise coupled to the pusher assembly **58**, the docking device **52** can advance in lockstep with the pusher assembly **58** through and/or out of the delivery shaft **54**.

[0068] In addition to the pusher shaft, in certain instances, the pusher assembly **58** can also include a sleeve shaft. The pusher shaft can be configured to advance the docking device **52** through the delivery shaft **54** and out of the distal end portion **53** of the delivery shaft **54**, while the sleeve shaft, when included, can have a distal dock sleeve configured to cover the docking device **52** within the delivery shaft **54** and while pushing the docking device **52** out of the delivery shaft **54** and positioning the docking device **52** at the implantation site. In some examples, the pusher shaft can be covered, at least in part, by the sleeve shaft. In certain examples, the sleeve shaft can include telescopic shaft members, as described further below.

[0069] In some examples, the pusher assembly **58** can comprise a pusher handle that is coupled to the pusher shaft and that is configured to be gripped and pushed by the user to translate the pusher shaft axially relative to the delivery shaft **54** (for example, to push the pusher shaft into and/or out of the distal end portion **53** of the delivery shaft **54**). The dock sleeve can be configured to be retracted and/or withdrawn from the docking device **52**, after positioning the docking device **52** at the target implantation site. For example, the pusher assembly **58** can include a sleeve handle that is coupled to the sleeve shaft and is configured to be pulled by a user to retract (for example, axially move) the sleeve shaft relative to the pusher shaft, thereby retracting the dock sleeve.

[0070] The pusher assembly **58** can be removably coupled to the docking device **52**, and as such can be configured to release, detach, decouple, and/or otherwise disconnect from the docking device **52** once the docking device **52** has been deployed at the target implantation site. As just one example, the pusher assembly **58** may be removably coupled to the docking device **52** via a thread, string, yarn, suture, or other suitable material that is tied or sutured to the docking device **52**.

[0071] In some examples, the pusher assembly **58** can include a suture lock assembly (also referred to as a “suture lock”) that is configured to receive and/or hold the thread or other suitable material that is coupled to the docking device **52** via a suture. The thread or other suitable material that forms the suture can extend from the docking device **52**, through the pusher assembly **58**, to the suture lock assembly. The suture lock assembly can also be configured to cut the suture to release, detach, decouple, and/or otherwise disconnect the docking device **52** from the pusher assembly **58**. For example, the suture lock assembly can comprise a cutting mechanism that is configured to be adjusted by the user to cut the suture.

[0072] Referring again to FIG. **2**A, after the guide catheter **30** is positioned within the left atrium **18**, the user may insert the docking device delivery apparatus **50** (for example, the delivery shaft **54**) into the patient **10** by advancing the delivery shaft **54** of the docking device delivery apparatus **50** through the guide catheter **30** and over the guidewire **40**. In some examples, the guidewire **40** can be at least partially retracted away from the left atrium **18** and into the guide catheter **30**. The user may then continue to advance the delivery shaft **54** of the docking device delivery apparatus **50** through the vasculature **12** along the guidewire **40** until the delivery shaft **54** reaches the left atrium **18**, as illustrated in FIG. **2**A. Specifically, the user may advance the delivery shaft **54** of the docking device delivery apparatus **50** by gripping and exerting a force on (for example, pushing) the handle **56** of the docking device delivery apparatus **50** toward the patient **10**. While advancing the delivery shaft **54** through the vasculature **12** and the heart **14**, the user may adjust the one or more articulation members **57** of the handle **56** to navigate the various turns, corners, constrictions, and/or other obstacles in the vasculature **12** and the heart **14**.

[0073] Once the delivery shaft **54** reaches the left atrium **18** and extends out of a distal end of the guide catheter **30**, the user can position the distal end portion **53** of the delivery shaft **54** at and/or near the posteromedial commissure of the native mitral valve **16** using the handle **56** (for example, the articulation members **57**). The user may then push the docking device **52** out of the distal end portion **53** of the delivery shaft **54** with the shaft of the pusher assembly **58** to deploy and/or implant the docking device **52** within the annulus of the native mitral valve **16**.

[0074] In some examples, the docking device **52** may be constructed from, formed of, and/or comprise a shape memory material, and as such, may return to its original, pre-formed shape when it exits the delivery shaft **54** and is no longer constrained by the delivery shaft **54**. As one example, the docking device **52** may originally be formed as a coil, and thus may wrap around leaflets **24** of the native mitral valve **16** as it exits the delivery shaft **54** and returns to its original coiled configuration.

[0075] After pushing a ventricular portion of the docking device **52** (for example, the portion of the docking device **52** shown in FIG. **2**A that is configured to be positioned within a left ventricle **26** and/or on the ventricular side of the native mitral valve **16**), the user may then deploy the remaining portion of the docking device **52** (for example, an atrial portion of the docking device **52**) from the delivery shaft **54** within the left atrium **18** by retracting the delivery shaft **54** away from the posteromedial commissure of the native mitral valve **16**. For example, the user can maintain the position of the pusher assembly **58** (for example, by exerting a holding and/or pushing force on the pusher shaft) while retracting the delivery shaft **54** proximally so that the delivery shaft **54** withdraws and/or otherwise retracts relative to the docking device **52** and the pusher assembly **58**. In this way, the pusher assembly **58** can hold the docking device **52** in place while the user retracts the delivery shaft **54**, thereby releasing the docking device **52** from the delivery shaft **54**. In some examples, the user can also remove the dock sleeve from the docking device **52**, for example, by retracting the sleeve shaft.

[0076] After deploying and implanting the docking device **52** at the native mitral valve **16**, the user may disconnect the docking device delivery apparatus **50** from the docking device **52**. Once the docking device **52** can be disconnected from the docking device delivery apparatus **50** (for example, by cutting the suture tied to the docking device **52**), the user may retract the docking device delivery apparatus **50** out of the vasculature **12** and away from the patient **10** so that the user can deliver and implant a prosthetic heart valve **62** within the implanted docking device **52** at the native mitral valve **16**.

[0077] FIG. **2**B depicts a third stage in the mitral valve replacement procedure, where the docking device **52** has been fully deployed and implanted at the native mitral valve **16** and the docking device delivery apparatus **50** (including the delivery shaft **54**) has been removed from the patient **10** such that only the guidewire **40** and the guide catheter **30** remain inside the patient **10**. In some examples, after removing the docking device delivery apparatus, the guidewire **40** can be advanced out of the guide catheter **30**, through the implanted docking device **52** at the native mitral valve **16**, and into the left ventricle **26** (FIG. **2**A). As such, the guidewire **40** can help to guide the prosthetic valve delivery apparatus **60** through the annulus of the native mitral valve **16** and at least partially into the left ventricle **26**.

[0078] As illustrated in FIG. **2**B, the docking device **52** can comprise a plurality of helical turns that wrap around the leaflets **24** of the native mitral valve **16** (within the left ventricle **26**). The implanted docking device **52** can have a more cylindrical shape than the annulus of the native mitral valve **16**, thereby providing a geometry that more closely matches the shape or profile of the prosthetic heart valve to be implanted. As a result, the docking device **52** can provide a tighter fit, and thus a better seal, between the prosthetic heart valve and the native mitral valve **16**, as described further below.

[0079] FIG. **3**A depicts a fourth stage in the mitral valve replacement procedure where the user is delivering and/or implanting a prosthetic heart valve **62** within the docking device **52** using a prosthetic valve delivery apparatus **60**.

[0080] As shown in FIG. **3**A, the prosthetic valve delivery apparatus **60** can comprise a delivery shaft **64** and a handle **66**. The delivery shaft **64** can extend distally from the handle **66**. The delivery shaft **64** can be configured to extend into the patient's vasculature **12** to deliver, implant, expand, and/or otherwise deploy the prosthetic heart valve **62** within the docking device **52** at the native mitral valve **16**. The handle **66** can be configured to be gripped and/or otherwise held by the user to advance the delivery shaft **64** through the patient's vasculature **12**.

[0081] In some examples, the handle **66** can comprise one or more articulation members **68** that are configured to aid in navigating the delivery shaft **64** through the vasculature **12** and the heart **14**. Specifically, the articulation members **68** can comprise one or more of knobs, buttons, wheels, and/or other types of physically adjustable control members that are configured to be adjusted by the user to flex, bend, twist, turn, and/or otherwise articulate a distal end portion of the delivery shaft **64** to aid in navigating the delivery shaft **64** through the vasculature **12** and into the left atrium **18** and left ventricle **26** of the heart **14**.

[0082] In some examples, the prosthetic valve delivery apparatus **60** can include an expansion mechanism **65** that is configured to radially expand and deploy the prosthetic heart valve **62** at the implantation site. In some instances, as shown in FIG. **3**A, the expansion mechanism **65** can comprise an inflatable balloon that is configured to be inflated to radially expand the prosthetic heart valve **62** within the docking device **52**. The inflatable balloon can be coupled to the distal end portion of the delivery shaft **64**.

[0083] In other examples, the prosthetic heart valve **62** can be self-expanding and can be configured to radially expand on its own upon removable of a sheath or capsule covering the radially compressed prosthetic heart valve **62** on the distal end portion of the delivery shaft **64**. In still other examples, the prosthetic heart valve **62** can be mechanically expandable and the prosthetic valve delivery apparatus **60** can include one or more mechanical actuators (for example, the expansion mechanism) configured to radially expand the prosthetic heart valve **62**.

[0084] As shown in FIG. **3**A, the prosthetic heart valve **62** can be mounted around the expansion mechanism **65** (for example, the inflatable balloon) on the distal end portion of the delivery shaft **64**, in a radially compressed configuration.

[0085] To navigate the distal end portion of the delivery shaft **64** to the implantation site, the user can insert the prosthetic valve delivery apparatus **60** (for example, the delivery shaft **64**) into the patient **10** through the guide catheter **30** and over the guidewire **40**. The user can continue to advance the prosthetic valve delivery apparatus **60** along the guidewire **40** (for example, through the vasculature **12**) until the distal end portion of the delivery shaft **64** reaches the native mitral valve **16**, as illustrated in FIG. **3**A. More specifically, the user can advance the delivery shaft **64** of the prosthetic valve delivery apparatus **60** by gripping and exerting a force on (for example, pushing) the handle **66**. While advancing the delivery shaft **64** through the vasculature **12** and the heart **14**, the user can adjust the one or more articulation members **68** of the handle **66** to navigate the various turns, corners, constrictions, and/or other obstacles in the vasculature **12** and heart **14**.

[0086] The user can advance the delivery shaft **64** along the guidewire **40** until the radially compressed prosthetic heart valve **62** mounted around the distal end portion of the delivery shaft **64** is positioned within the docking device **52** and the native mitral valve **16**. In some examples, as shown in FIG. **3**A, a distal end of the delivery shaft **64** and a least a portion of the radially compressed prosthetic heart valve **62** can be positioned within the left ventricle **26**.

[0087] Once the radially compressed prosthetic heart valve **62** is appropriately positioned within the docking device **52** (FIG. **3**A), the user can manipulate one or more actuation mechanisms of the handle **66** of the prosthetic valve delivery apparatus **60** to actuate the expansion mechanism **65** (for example, inflate the inflatable balloon), thereby radially expanding the prosthetic heart valve **62** within the docking device **52**. In some examples, the user can lock the prosthetic heart valve **62** in its fully expanded position (for example, with a locking mechanism) to prevent the prosthetic heart valve **62** from collapsing.

[0088] FIG. **3**B shows a fifth stage in the mitral valve replacement procedure where the prosthetic heart valve **62** in its radially expanded configuration and implanted within the docking device **52** in the native mitral valve **16**. As shown in FIG. **3**B, the prosthetic heart valve **62** can be received and retained within the docking device **52**.

[0089] As also shown in FIG. **3**B, after the prosthetic heart valve **62** has been fully deployed and implanted within the docking device **52** at the native mitral valve **16**, the prosthetic valve delivery apparatus **60** (including the delivery shaft **64**) can be removed from the patient **10** such that only the guidewire **40** and the guide catheter **30** remain inside the patient **10**.

[0090] FIG. **4** depicts a sixth stage in the mitral valve replacement procedure, where the guidewire **40** and the guide catheter **30** have been removed from the patient **10**. The docking device **52** can be configured to provide a seal between the prosthetic heart valve **62** and the leaflets **24** of the native mitral valve **16** to reduce paravalvular leakage around the prosthetic heart valve **62**. Specifically, the docking device **52** can initially constrict the leaflets **24** of the native mitral valve **16**. The prosthetic heart valve **62** can then push the leaflets **24** against the docking device **52** as it radially expands within the docking device **52**. Thus, the docking device **52** and the prosthetic heart valve **62** can be configured to sandwich the leaflets **24** of the native mitral valve **16** when the prosthetic heart valve **62** is expanded within the docking device **52**. In this way, the docking device **52** can provide a seal between the leaflets **24** of the native mitral valve **16** and the prosthetic heart valve **62** to reduce paravalvular leakage around the prosthetic heart valve **62**.

[0091] In some examples, one or more of the docking device delivery apparatus **50**, the prosthetic valve delivery apparatus **60**, and/or the guide catheter **30** can comprise one or more fluid ports that are configured to supply flushing fluid to the lumens thereof to prevent and/or reduce the likelihood of blood clot (for example, thrombus) formation. Example fluid ports that can be used to inject flushing fluid into a docking device delivery apparatus are described further below.

[0092] Although FIGS. **1-4** specifically depict a mitral valve replacement procedure, it should be appreciated that the same and/or similar procedure may be utilized to replace other heart valves (for example, tricuspid, pulmonary, and/or aortic valves). Further, the same and/or similar delivery apparatuses (for example, docking device delivery apparatus **50**, prosthetic valve delivery apparatus **60**, guide catheter **30**, and/or guidewire **40**), docking devices (for example, docking device **52**), replacement heart valves (for example, prosthetic heart valve **62**), and/or components thereof may be utilized for replacing these other heart valves.

[0093] For example, when replacing a native tricuspid valve, the user may also access the right atrium **20** via a femoral vein but may not need to cross the atrial septum **22** into the left atrium **18**. Instead, the user may leave the guidewire **40** in the right atrium **20** and perform the same and/or similar docking device implantation process at the tricuspid valve. Specifically, the user may push the docking device **52** out of the delivery shaft **54** around the ventricular side of the tricuspid valve leaflets, release the remaining portion of the docking device **52** from the delivery shaft **54** within the right atrium **20**, and then remove the delivery shaft **54** of the docking device delivery apparatus **50** from the patient **10**. The user may then advance the guidewire **40** through the tricuspid valve into the right ventricle and perform the same and/or similar prosthetic heart valve implantation process at the tricuspid valve, within the docking device **52**. Specifically, the user may advance the delivery shaft **64** of the prosthetic valve delivery apparatus **60** through the patient's vasculature along the guidewire **40** until the prosthetic heart valve **62** is positioned or disposed within the docking device **52** and the tricuspid valve. The user may then expand the prosthetic heart valve **62** within the docking device **52** before removing the prosthetic valve delivery apparatus **60** from the patient **10**. In another example, the user may perform the same and/or similar process to replace the aortic valve but may access the aortic valve from the outflow side of the aortic valve via a femoral artery.

[0094] Further, although FIGS. **1-4** depict a mitral valve replacement procedure that accesses the native mitral valve **16** from the left atrium **18** via the right atrium **20** and femoral vein, it should be appreciated that the native mitral valve **16** may alternatively be accessed from the left ventricle **26**. For example, the user may access the native mitral valve **16** from the left ventricle **26** via the aortic valve by advancing one or more delivery apparatuses through an artery to the aortic valve, and then through the aortic valve into the left ventricle **26**.

[0095] Additional examples of the docking device delivery apparatus, including its variants, and methods of implanting a docking device and implanting a prosthetic valve within the docking device are described in PCT Patent Application Publication Nos. WO 2020/247907 and WO 2022/087336, and U.S. Patent Publication Nos. US2018/0318079, US2018/0263764, and US2018/0177594, which are all incorporated by reference herein in their entireties.

Exemplary Docking Devices

[0096] Docking devices according to examples of the disclosure can, for example, provide a stable anchoring site, landing zone, or implantation zone at the implant site in which prosthetic valves can be expanded or otherwise implanted. Many of the disclosed docking devices comprise a circular or cylindrically-shaped portion, which can (for example) allow a prosthetic heart valve comprising a circular or cylindrically-shaped valve frame to be expanded or otherwise implanted into native locations with naturally circular cross-sectional profiles and/or in native locations with naturally with non-circular cross sections. In addition to providing an anchoring site for the prosthetic valve, the docking devices can be sized and shaped to cinch or draw the native valve (for example, mitral, tricuspid, etc.) anatomy radially inwards. In this manner, one of the main causes of valve regurgitation (for example, functional mitral regurgitation), specifically enlargement of the heart (for example, enlargement of the left ventricle, etc.) and/or valve annulus, and consequent stretching out of the native valve (for example, mitral, etc.) annulus, can be at least partially offset or counteracted. Some examples of the docking devices further include features which, for example, are shaped and/or modified to better hold a position or shape of the docking device during and/or after expansion of a prosthetic valve therein. By providing such docking devices, replacement valves can be more securely implanted and held at various valve annuluses, including at the mitral valve annulus which does not have a naturally circular cross-section.

[0097] In some instances, a docking device can comprise a paravalvular leakage (PVL) guard (also referred to herein as "a guard member"). The PVL guard can, for example, help reduce regurgitation and/or promote tissue ingrowth between the native tissue and the docking device, thereby reducing or preventing migration of the docking device (and the prosthetic valve received within the docking device) relative to the native tissue.

[0098] The PVL guard can, in some examples, be movable between a delivery configuration and a deployed configuration. When the PVL guard is in the delivery configuration, an outer edge of the PVL guard can extend along and adjacent the coil. When the PVL guard is in the deployed configuration, the outer edge of the PVL guard can form a helical shape rotating about a central longitudinal axis of the coil and at least a segment of the outer edge of PVL guard can extend radially away from the coil.

[0099] In certain examples, the PVL guard can cover or surround a portion of a coil of the docking device. As described more fully below, such PVL guard can move from a radially compressed (and axially elongated) state to a radially expanded (and axially foreshortened) state, and a proximal end portion of the PVL guard can be axially movable relative to the coil.

[0100] FIGS. **5**A-**5**D show a docking device **100**, according to one example. The docking device **100** can, for example, be implanted within a native valve annulus. As depicted in FIGS. **7**A-**7**B, the docking device can be configured to receive and secure a prosthetic valve within the docking device, thereby securing the prosthetic valve at the native valve annulus.

[0101] Referring to FIGS. **5**A-**5**D, the docking device **100** can comprise a coil **102** and a guard member **104** covering at least a portion of the coil **102**. In certain examples, the coil **102** can include a shape memory material (for example, nickel titanium alloy or "Nitinol") such that the docking device **100** (and the coil **102**) can move from a substantially straight configuration (also referred to as "delivery configuration") when disposed within a delivery sheath of a delivery apparatus to a helical configuration (also referred to as "deployed configuration," as shown in FIGS. **5**A-**5**B) after being removed from the delivery sheath.

[0102] In certain examples, when the guard member **104** is in the deployed configuration, the guard member **104** can extend circumferentially relative to a central longitudinal axis **101** of the docking device **100** from 180 degrees to 400 degrees, or from 210 degrees to 330 degrees, or from 250 degrees to 290 degrees, or from 260 degrees to 280 degrees. In one particularly example, when the guard member **104** is in the deployed configuration, the guard member **104** can extend circumferentially 270 degrees relative to the central longitudinal axis **101**. In other words, the guard member **104** can extend circumferentially from about one half of a revolution (for example, 180 degrees) around the central longitudinal axis **101** in some examples to more than a full revolution (for example, 400 degrees) around the central longitudinal axis **101** in other examples, including various ranges in between. As used herein, a range (for example, 180-400 degrees, from 180 degrees to 400 degrees, and between 180 degrees and 400 degrees) includes the endpoints of the range (for example, 180 degrees and 400 degrees).

[0103] In some examples, the docking device **100** can also include a retention member **114** surrounding at least a portion of the coil **102** and at least being partially covered by the guard member **104**. In some examples, the retention member **114** can comprise a braided material. In some examples, the retention member **114** can include a woven material. In addition, the retention member **114** can provide a surface area that encourages or promotes tissue ingrowth and/or adherence, and/or reduce trauma to native tissue. For example, in certain instances, the retention member **114** can have a textured outer surface configured to promote tissue ingrowth. In certain instances, the retention member **114** can be impregnated with growth factors to stimulate or promote tissue ingrowth.

[0104] In some examples, as illustrated in FIGS. **5**A-**5**B and **7**A-**7**B, at least a proximal end portion of the retention member **114** can extend out of a proximal end of the guard member **104**. In some examples, at least a distal end portion of the retention member **114** can extend out of a distal end of the guard member **104**. In one example, the retention member **114** can be completely covered by the guard member **104**.

[0105] In some examples, the retention member **114** can be configured to interact with the guard member **104** to limit or resist motion of the guard member **104** relative to the coil **102**. For example, a proximal end **105** of the guard member **104** can have an inner diameter that is about the same as an outer diameter of the retention member **114**. As such, an inner surface of the guard member **104** at the proximal end **105** can frictionally interact or engage with the retention member **114** so that axial movement of the proximal end **105** of the guard member **104** relative to the coil **102** can be impeded by a frictional force exerted by the retention member **114**.

[0106] The coil **102** has a proximal end **102***p* and a distal end **102***d* (which also respectively define the proximal and distal ends of the docking device **100**). When being disposed within the delivery sheath (for example, during delivery of the docking device into the vasculature of a patient), a body of the coil **102** between the proximal end **102***p* and distal end **102***d* can form a generally straight delivery configuration (that is, without any coiled or looped portions, but can be flexed or bent) so as to maintain a small radial profile when moving through a patient's vasculature. After being removed from the delivery sheath and deployed at an implant position, the coil **102** can move from the delivery configuration to the helical deployed configuration and wrap around native tissue adjacent the implant position. For example, when implanting the docking device at the location of a native valve, the coil **102** can be configured to surround native leaflets of the native valve (and the chordae tendineae that connects native leaflets to adjacent papillary muscles, if present), as described above.

[0107] The docking device **100** can be releasably coupled to a delivery apparatus. For example, in certain examples, the docking device **100** can be coupled to a delivery apparatus via a release suture that can be configured to be tied to the docking device **100** and cut for removal. In one example, the release suture can be tied to the docking device **100** through an eyelet or eyehole **103** located adjacent the proximal end **102***p* of the coil. In another example, the release suture can be tied around a circumferential recess that is located adjacent the proximal end **102***p* of the coil **102**.

[0108] In some examples, the docking device **100** in the deployed configuration can be configured to fit at the mitral valve position. In other examples, the docking device can also be shaped and/or adapted for implantation at other native valve positions as well, such as at the tricuspid valve. As described herein, the geometry of the docking device **100** can be configured to engage the native anatomy, which can, for example, provide for increased stability and reduction of relative motion between the docking device **100**, the prosthetic valve docked therein, and/or the native anatomy. Reduction of such relative motion can, among other things, prevent material degradation of components of the docking device **100** and/or the prosthetic valve docked therein and/or prevent damage or trauma to the native tissue.

[0109] As shown in FIGS. **5**A-**5**B, the coil **102** in the deployed configuration can include a leading turn **106** (or "leading coil"), a central region **108**, and a stabilization turn **110** (or "stabilization coil") around the central longitudinal axis **101**. The central region **108** can possess one or more helical turns having substantially equal inner diameters. The leading turn **106** can extend from a distal end of the central region **108** and has a diameter greater than the diameter of the central region **108** (in one or more configurations). The stabilization turn **110** can extend from a proximal end of the central region **108** and has a diameter greater than the diameter of the central region **108** (in one or more configurations).

[0110] In certain examples, the central region **108** can include a plurality of helical turns, such as a proximal turn **108***p* in connection with the stabilization turn **110**, a distal turn **108***d* in connection with the leading turn **106**, and one or more intermediate turns **108***m* disposed between the proximal turn **108***p* and the distal turn **108***d*. In the example shown in FIG. **5**A, there is only one intermediate turn **108***m* between the proximal turn **108***p* and the distal turn **108***d*. In other examples, there are more than one intermediate turns **108***m* between the proximal turn **108***p* and the distal turn **108***d*. Some of the helical turns in the central region **108** can be full turns (that is, rotating 360 degrees). In some examples, the proximal turn **108***p* and/or the distal turn **108***d* can be partial turns (for example, rotating less than 360 degrees, such as 180 degrees, 270 degrees, etc.).

[0111] A size of the docking device **100** can be generally selected based on the size of the desired prosthetic valve to be implanted into the patient. In certain examples, the central region **108** can be configured to retain a radially expandable prosthetic valve (as shown in FIGS. **7**A-**7**B). For example, the inner diameter of the helical turns in the central region **108** can be configured to be smaller than an outer diameter of the prosthetic valve when the prosthetic valve is radially expanded so that additional radial force can act between the central region **108** and the prosthetic valve to hold the prosthetic valve in place. The helical turns (for example, **108***p*, **108***m*, **108***d*) in the central region **108** can also be referred to herein as "functional turns."

[0112] The stabilization turn **110** can be configured to help stabilize the docking device **100** in the desired position. For example, the radial dimension of the stabilization turn **110** can be significantly larger than the radial dimension of the coil in the central region **108**, so that the stabilization turn **110** can flare or extend sufficiently outwardly so as to abut or push against the walls of the circulatory system, thereby improving the ability of the docking device **100** to stay in its desired position prior to the implantation of the prosthetic valve. In some examples, the diameter of stabilization turn **110** is desirably larger than the native annulus, native valve plane, and/or native chamber for better stabilization. In some examples, the stabilization turn **110** can be a full turn (that is, rotating about 360 degrees). In some examples, the stabilization turn **110** can be a partial turn (for example, rotating between about 180 degrees and about 270 degrees).

[0113] In one particularly example, when implanting the docking device **100** at the native mitral valve location, the functional turns in the central region **108** can be disposed substantially in the left ventricle and the stabilization turn **110** can be disposed substantially in the left atrium. The stabilization turn **110** can be configured to provide one or more points or regions of contact between the docking device **100** and the left atrial wall, such as at least three points of contact in the left atrium or complete contact on the left atrial wall. In certain examples, the points of contact between the docking device **100** and the left atrial wall can form a plane that is approximately parallel to a plane of the native mitral valve.

[0114] In some examples, the stabilization turn **110** can have an atrial portion **110***a* in connection with the proximal turn **108***p* of the central region **108**, a stabilization portion **110***c* adjacent to the proximal end **102***p* of the coil **102**, and an ascending portion **110***b* located between the atrial portion **110***a* and the stabilization portion **110***c*. Both the atrial portion **110***a* and the stabilization portion **110***c* can be generally parallel to the helical turns in the central region **108**, whereas the ascending portion **110***b* can be oriented to be angular relative to the atrial portion **110***a* and the stabilization portion **110***c*. For example, in certain examples, the ascending portion **110***b* and the stabilization portion **110***c* can form an angle from about 45 degrees to about 90 degrees (inclusive). In certain examples, the stabilization portion **110***c* can define a plane that is substantially parallel to a plane defined by the atrial portion **110***a*. A boundary **107** (marked by a dashed line in FIG. **5**A) between the ascending portion **110***b* and the stabilization portion **110***c* can be determined as a location where the ascending portion **110***b* intersects the plane defined by the stabilization portion **110***c*. The curvature of the stabilization turn **110** can be configured so that the atrial portion **110***a* and the stabilization portion **110***c* are disposed on approximately opposite sides when the docking device **100** is fully expanded. When implanting the docking device **100** at the native mitral valve location, the atrial portion **110***a* can be configured to abut the posterior wall of the left atrium and the stabilization portion **110***c* can be configured to flare out and press against the anterior wall of the left atrium.

[0115] As noted above, the leading turn **106** can have a larger radial dimension than the helical turns in the central region **108**. As described herein, the leading turn **106** can help more easily guide the coil **102** around and/or through the chordae tendineae and/or adequately around all native leaflets of the native valve (for example, the native mitral valve, tricuspid valve, etc.). For example, once the leading turn **106** is navigated around the desired native anatomy, the remaining coil (such as the functional turns) of the docking device **100** can also be guided around the same features. In some examples, the leading turn **106** can be a full turn (that is, rotating about 360 degrees). In some examples, the leading turn **106** can be a partial turn (for example, rotating between about 180 degrees and about 270 degrees). When a prosthetic valve is radially expanded within the central region **108** of the coil, the functional turns in the central region **108** can be further radially expanded. As a result, the leading turn **106** can be pulled in the proximal direction and become a part of the functional turn in the central region **108**.

[0116] In certain examples, at least a portion of the coil **102** can be surrounded by a first cover **112**. As shown in FIGS. **5**C-**5**D, the first cover **112** can have a tubular shape and thus can also be referred to as a "tubular member." In certain examples, the tubular member **112** can cover an entire length of the coil **102**. In certain examples, the tubular member **112** covers only selected portion(s) of the coil **102**.

[0117] In certain examples, the tubular member **112** can be coated on and/or bonded on the coil **102**. In certain examples, the tubular member **112** can be a cushioned, padded-type layer protecting the coil. The tubular member **112** can be constructed of various native and/or synthetic materials. In one particularly example, the tubular member **112** can include expanded polytetrafluoroethylene (ePTFE). In certain examples, the tubular member **112** is configured to be fixedly attached to the coil **102** (for example, by means of textured surface resistance, suture, glue, thermal bonding, or any other means) so that relative axial movement between the tubular member **112** and the coil **102** is restricted or prohibited.

[0118] In some examples, as illustrated in FIGS. **5**C-**5**D, at least a portion of the tubular member **112** is surrounded by the retention member **114**. In some examples, the tubular member **112** can extend through an entire length of the retention member **114**. In some examples, at least a portion of the tubular member **112** may not be surrounded by the retention member **114**.

[0119] In some examples, a distal end portion of the retention member **114** can extent axially beyond (that is, positioned distal to) the distal end of the guard member **104**, and a proximal end portion of the retention member **114** can extend axially beyond (that is, positioned proximal to) the proximal end **105** of the guard member **104** to aid retention of prosthetic valve and tissue ingrowth. In some examples, a distal end of the retention member **114** can be positioned adjacent the leading turn **106** (for example, near the location marked by the dashed line **109** in FIG. **5**A). For example, the retention member **114** can cover the functional turns of the coil **102** in the central region **108**. Thus, when the docking device **100** is deployed at the native valve and the prosthetic valve is radially expanded within the docking device **100**, the retention member **114** at the central region **108** can frictionally engage the prosthetic valve. In another example, the distal end of the retention member **114** can be disposed at or adjacent the distal end **102***d* of the coil **102**. In the example depicted in FIGS. **5**A-**5**B, a proximal end of the retention member **114** can be disposed at or adjacent the ascending portion **110***b* of the coil **102**.

[0120] In certain examples, the docking device **100** can have one or more seating markers. For example, FIGS. **5**A-**5**B show a proximal seating marker **121***p* and a distal seating marker **121***d*, wherein the proximal seating marker **121***p* is positioned proximal relative to the distal seating marker **121***d*. Both the proximal and distal seating markers **121***p*, **121***d* can have predefined locations relative to the coil **102**. As shown, both the proximal and distal seating markers **121***p*, **121***d* can be disposed distal to the ascending portion **110***b*, for example, at the atrial portion **110***a*, of the coil **102**.

[0121] In certain examples, both the proximal and distal seating markers **121***p*, **121***d* can include a radiopaque material so that these seating markers can be visible under fluoroscopy such as during an implantation procedure. The seating markers **121***p*, **121***d* can be used to mark the proximal and distal boundaries of a segment of the coil **102** where the proximal end **105** of the guard member **104** can be positioned when deploying the docking device **100**.

[0122] In certain examples, the seating markers **121***p*, **121***d* can be disposed on the tubular member **112** and covered by the retention member **114**. In some examples, the seating markers **121***p*, **121***d* can be disposed on the atrial portion **110***a* of the coil **102** and covered by the tubular member **112**. In particularly examples, the seating markers **121***p*, **121***d* can be disposed directly on the retention member **114**. In yet alternative examples, the seating markers **121***p*, **121***d* can be disposed on different layers relative to each other. For example, one of the seating markers (for example, **121***p*) can be disposed outside the tubular member **112** and covered by the retention member **114**, whereas another seating marker (for example, **121***d*) can be disposed directly on the coil **102** and covered by the tubular member **112**.

[0123] In certain examples, a segment of the coil **102** located between the proximal seating marker **121***p* and the distal seating marker **121***d* can have an axial length between about 2 mm and about 7 mm, or between about 3 mm and about 5 mm. In one specific example, the axial length of the coil segment between the proximal seating marker **121***p* and the distal seating marker **121***d* is about 4 mm.

[0124] In certain examples, an axial distance between the proximal seating marker **121***p* and a distal end of the ascending portion **110***b* is between about 10 mm and about 30 mm, or between about 15 mm and about 25 mm. In one specific example, the axial distance between the proximal seating marker **121***p* and the distal end of the ascending portion **110***b* is about 20 mm.

[0125] Although two seating markers **121***p*, **121***d* are shown in FIGS. **5**A-**5**B, it is to be understood that the number of seating markers can be more than two or less than two. For example, in one example, the docking device **100** can have only one seating marker (for example, **121***p*). In another example, one or more additional seating markers can be placed between the proximal and distal seating markers **121***p*, **121***d*. As noted above, the proximal end **105** of the guard member can be positioned between the proximal and distal seating markers **121***p*, **121***d* when deploying the docking device **100**. As such, these additional seating markers can function as a scale to indicate a precise location of the proximal end **105** of the guard member **104** relative to the coil **102**.

[0126] As described herein, the guard member **104** can constitute a part of a cover assembly **120** for the docking device **100**. In some examples, the cover assembly **120** can also include the tubular member **112**. In some examples, the cover assembly **120** can further include the retention member **114**.

[0127] In some examples, as shown in FIGS. **5**A-**5**B, when the docking device **100** is in the deployed configuration, the guard member **104** can be configured to cover a portion (for example, the atrial portion **110***a*) of the stabilization turn **110** of the coil **102**. In certain examples, the guard member **104** can be configured to cover at least a portion of the central region **108** of the coil **102**, such as a portion of the proximal turn **108***p*. In certain examples, the guard member **104** can extend over the entirety of the coil **102**.

[0128] As described herein, the guard member **104** can radially expand so as to help preventing and/or reducing paravalvular leakage. Specifically, the guard member **104** can be configured to radially expand such that an improved seal is formed closer to and/or against a prosthetic valve deployed within the docking device **100**. In some examples, the guard member **104** can be configured to prevent and/or inhibit leakage at the location where the docking device **100** crosses between leaflets of the native valve (for example, at the commissures of the native leaflets). For example, without the guard member **104**, the docking device **100** may push the native leaflets apart at the point of crossing the native leaflets and allow for leakage at that point (for example, along the docking device or to its sides). However, the guard member **104** can be configured to expand to cover and/or fill any opening at that point and inhibit leakage along the docking device **100**.

[0129] In another example, when the docking device **100** is deployed at a native atrioventricular valve, the guard member **104** covers predominantly a portion of the stabilization turn **110** and/or a portion of the central region **108**. In one example, the guard member **104** can cover predominantly the atrial portion **110***a* of the stabilization turn **110** that is located distal to the ascending portion **110***b*. Thus, the guard member **104** does not extend into the ascending portion **110***b* (or at least the guard member **104** can terminate before the anterolateral commissure of the native valve) when the docking device **100** is in the deployed configuration. In certain circumstances, the guard member **104** can extend onto the ascending portion **110***b*. This may cause the guard member **104** to kink, which (in some instances) may reduce the performance and/or durability of the guard member. Thus, the retention member **114** can, among other things, improve the functionality and/or longevity of the guard member **104** by preventing the guard member **104** from extending into the ascending portion **110***b* of the coil **102**.

[0130] Yet in alternative examples, the guard member **104** can cover not only the atrial portion **110***a*, but can also extend over the ascending portion **110***b* of the stabilization turn **110**. This can occur, for example, in circumstances when the docking device is implanted in other anatomical locations and/or the guard member **104** is reinforced to reduce the risk of wire break.

[0131] In various examples, the guard member **104** can help covering an atrial side of an atrioventricular valve to prevent and/or inhibit blood from leaking through the native leaflets, commissures, and/or around an outside of the prosthetic valve by blocking blood in the atrium from flowing in an atrial to ventricular direction (that is, antegrade blood flow)—other than through the prosthetic valve. Positioning the guard member **104** on the atrial side of the valve can additionally or alternatively help reduce blood in the ventricle from flowing in a ventricular to atrial direction (that is, retrograde blood flow).

[0132] In some examples, the guard member **104** can be positioned on a ventricular side of an atrioventricular valve to prevent and/or inhibit blood from leaking through the native leaflets, commissures, and/or around an outside of the prosthetic valve by blocking blood in the ventricle from flowing in a ventricular to atrial direction (that is, retrograde blood flow). Positioning the guard member **104** on the ventricular side of the valve can additionally or alternatively help reduce blood in the atrium from flowing in the atrial direction to ventricular direction (that is, antegrade blood flow)—other than through the prosthetic valve.

[0133] The guard member **104** can include an expandable member **116** and a cover member **118** (also referred to as a “second cover” or an “outer cover”) surrounding an outer surface of the expandable member **116**. In certain examples, the expandable member **116** surrounds at least a portion of the tubular member **112**. In certain examples, the tubular member **112** can extend (completely or partially) through the expandable member **116**.

[0134] The expandable member **116** can extend radially outwardly from the coil **102** (and the tubular member **112**) and is movable between a radially compressed (and axially elongated) state and a radially expanded (and axially foreshortened) state. That is, the expandable member **116** can axially foreshorten when it moves from the radially compressed state to the radially expanded state and can axially elongate when it moves from the radially expanded state to the radially compressed state.

[0135] In certain examples, the expandable member **116** can include a braided structure, such as a braided wire mesh or lattice. In certain examples, the expandable member **116** can include a shape memory material that is shape set and/or pre-configured to expand to a particular shape and/or size when unconstrained (for example, when deployed at a native valve location). For example, the expandable member **116** can have a braided structure containing a shape memory alloy with Superelastic properties, such as Nitinol. In certain examples, the expandable member **116** can have a braided structure containing a ternary shape memory alloy with Superelastic properties, such as NiTiX where X can be chromium (Cr), cobalt (Co), zirconium (Zr), hafnium (Hf), etc. In certain examples, the expandable member **116** can comprise a metallic material that does not have the shape memory properties. Examples of such metallic material include cobalt-chromium, stainless steel, etc. In one specific example, the expandable member **116** can comprise nickel-free austenitic stainless steel in which nickel can be completely replaced by nitrogen. In another specific example, the expandable member **116** can comprise cobalt-chromium or cobalt-nickel-chromium-molybdenum alloy with significantly low density of titanium. The number of wires (or fibers, strands, or the like) forming the braided structure can be selected to achieve a desired elasticity and/or strength of the expandable member **116**. In certain examples, the number of wires used to braid the expandable member **116** can range from 16 to 128 (for example, 32 wires, 48 wires, 64 wires, 96 wires, etc.). In certain examples, the braid density can range from 20 picks per inch (PPI) to 70 PPI, or from 25 PPI to 65 PPI. In one specific example, the braid density is about 36 PPI. In another specific example, the braid density is about 40 PPI. In certain examples, the diameter of the wires can range from about 0.002 inch to about 0.004 inch. In one particularly example, the diameter of the wires can be about 0.003 inch. In another example, the expandable member **116** can be a combination of braided wire (which can include a shape memory material or non-shape memory material) and a polymeric material and/or textile (for example, polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), polyether ether ketone (PEEK), thermoplastic polyurethane (TPU), etc.). For example, the expandable member **116** can include a braided wireframe embedded in a polymeric material.

[0136] In some examples, the expandable member **116** can include a braided metallic wireframe coated with an elastomer (for example, ePTFE, TPU, or the like), which can elastically deform as the braided wireframe expands and/or compresses. In some examples, the expandable member **116** can comprise a braid and/or weave that includes one or more metallic wires and one or more polymeric fibers. In other words, the metallic wires and the polymeric fibers can be interwoven together to define a braided structure. In some instances, the polymeric fibers can have the same or about the same diameter as the metallic wires. In other instances, the polymeric fibers can have a smaller diameter (for example, microfibers) than the metallic wires, or vice versa.

[0137] In yet another example, the expandable member **116** can include a polymeric material, such as a thermoplastic material (for example, PET, polyether ether ketone (PEEK), thermoplastic polyurethane (TPU), etc.), without a braided wireframe.

[0138] In certain examples, the expandable member **116** can include a foam structure. For example, the expandable member can include an expandable memory foam which can expand to a specific shape or specific pre-set shape upon removal of a crimping pressure (for example, removal of the docking device **100** from the delivery sheath) prior to delivery of the docking device.

[0139] As described herein, the cover member **118** can be configured to be so elastic that when the expandable member **116** moves from the radially compressed (and axially elongated) state to the radially expanded (and axially foreshortened) state, the cover member **118** can also radially expand and axially foreshorten together with the expandable member **116**. In other words, the guard member **104**, as a whole, can move from a radially compressed (and axially elongated) state to a radially expanded (and axially foreshortened) state. As described herein, the radially expanded (and axially foreshortened) state is also referred to as the “relaxed state,” and the radially compressed (and axially elongated) state is also referred to as the “collapsed state.”

[0140] In certain examples, the cover member **118** can be configured to be atraumatic to native tissue and/or promote tissue ingrowth into the cover member **118**. For example, the cover member **118** can have pores to encourage tissue ingrowth. In another example, the cover member **118** can be impregnated with growth factors to stimulate or promote tissue ingrowth, such as transforming growth factor alpha (TGF-alpha), transforming growth factor beta (TGF-beta), basic fibroblast growth factor (bFGF), vascular epithelial growth factor (VEGF), and combinations thereof. The cover member **118** can be constructed of any suitable material, including foam, cloth, fabric, and/or polymer, which is flexible to allow for compression and expansion of the cover member **118**. In one example, the cover member **118** can include a fabric layer constructed from a thermoplastic polymer material, such as polyethylene terephthalate (PET).

[0141] As described herein, a distal end portion **104***d* of the guard member **104** (including a distal end portion of the expandable member **116** and a distal end portion of the cover member **118**) can be fixedly coupled to the coil **102** (for example, via suturing, gluing, or the like), and a proximal end portion **104***p* of the guard member **104** (including a proximal end portion of the expandable member **116** and a proximal end portion of the cover member **118**) can be axially movable relative to the coil **102**. Further, the proximal end portion of the expandable member **116** can be fixedly coupled to the proximal end portion of the cover member **118** (for example, via suturing, gluing, thermal compression, laser fusion, etc.).

[0142] Alternatively, the proximal end portion **104***p* of the guard member **104** can be fixedly coupled to the coil **102**, while a distal end portion **104***d* of the guard member **104** can be axially movable relative to the coil **102**.

[0143] When the docking device **100** is retained within the delivery sheath in the substantially straight configuration, the expandable member **116** can be radially compressed by the delivery sheath and remains in the radially compressed (and axially elongated) state. The radially compressed (and axially elongated) expandable member **116** can contact the retention member **114** (FIG. **5**D) so that no gap or cavity exists between the retention member **114** and the expandable member **116**.

[0144] After the docking device **100** is removed from the delivery sheath and changes from the delivery configuration to the deployed configuration, the guard member **104** can also move from a delivery configuration to a deployed configuration. In certain examples, a dock sleeve can be configured to cover and retain the docking device **100** within the delivery sheath when navigating the delivery sheath through the patient's native valve. The docking sleeve can also, for example, help to guide the docking device around the native leaflets and chordae. Retraction of the dock sleeve relative to the docking device **100** can expose the guard member **104** and cause it to move from the delivery configuration to the deployed configuration. Specifically, without the constraint of the delivery sheath and the dock sleeve, the expandable member **116** can radially expand (and axially foreshorten) so that a gap or cavity **111** can be created between the retention member **114** and the expandable member **116** (FIG. **5**C). Thus, when the guard member **104** is in the delivery configuration, an outer edge of the guard member **104** can extend along and adjacent the coil **102** (since there is no gap **111**, only the retention member **114** and/or the tubular member **112** separate the coil **102** from the expandable member **116**, as shown in FIG. **5**D). When the guard member **104** is in the deployed configuration, the outer edge of the guard member **104** can form a helical shape rotating about the central longitudinal axis **101** (FIGS. **5**A-**5**B and **7**A-**7**B) and at least a segment of the outer edge of guard member can extend radially away from the coil **102** (for example, due to the creation of the gap **111** between the expandable member **116** and the retention member **114**).

[0145] Because the distal end portion **104***d* of the guard member **104** is fixedly coupled to the coil **102** and the proximal end portion **104***p* of the guard member **104** can be axially movable relative to the coil **102**, the proximal end portion **104***p* of the guard member **104** can slide axially over the tubular member **112** and toward the distal end **102***d* of the coil **102** when expandable member **116** moves from the radially compressed state to the radially expanded state. As a result, the proximal end portion **104***p* of the guard member **104** can be disposed closer to the proximal end **102***p* of the coil **102** when the expandable member **116** is in the radially compressed state than in the radially expanded state.

[0146] In certain examples, the cover member **118** can be configured to engage with the prosthetic valve deployed within the docking device **100** so as to form a seal and reduce paravalvular leakage between the prosthetic valve and the docking device **100** when the expandable member **116** is in the radially expanded state. The cover member **118** can also be configured to engage with the native tissue (for example, the native annulus and/or native leaflets) to reduce PVL between the docking device and/or the prosthetic valve and the native tissue.

[0147] In certain examples, when the expandable member **116** is in the radially expanded state, the proximal end portion **104***p* of the guard member **104** can have a tapered shape as shown in FIGS.

5A-**5**B, such that the diameter of the proximal end portion **104***p* gradually increases from a proximal end **105** of the guard member **104** to a distally located body portion of the guard member **104**. This can, for example, help to facilitate loading the docking device into a delivery sheath of the delivery apparatus and/or retrieval and/or re-positioning of the docking device into the delivery apparatus during an implantation procedure. In addition, due to its small diameter, the proximal end **105** of the guard member **104** can frictionally engage with the retention member **114** so that the retention member **114** can reduce or prevent axial movement of the proximal end portion **104***p* of the guard member **104** relative to the coil **102**.

[0148] In certain examples, the docking device **100** can include at least one radiopaque marker configured to provide visual indication about the location of the docking device **100** relative to its surrounding anatomy, and/or the amount of radial expansion of the docking device **100** (for example, when a prosthetic valve is subsequently deployed in the docking device **100**) under fluoroscopy. For example, one or more radiopaque markers can be placed on the coil **102**. In one particularly example, a radiopaque marker (which can be larger than the seating markers **121***p*, **121***d*) can be disposed at the central region **108** of the coil. In another example, one or more radiopaque markers can be placed on the tubular member **112**, the expandable member **116**, and/or the cover member **118**. As noted above, the docking device **100** can also have one or more radiopaque markers (for example, **121***p* and/or **121***d*) located distal to the ascending portion **110***b* of the coil **102**. The radiopaque marker(s) used to provide visual indication about the location and/or the amount of radial expansion of the docking device **100** can be in addition to the seating markers (for example, **121***p*, **121***d*) described above.

[0149] Further details of the docking device and its variants, including various examples of the coil, the first cover (or tubular member), the second cover (or cover member), the expandable member, and other components of the docking device, are described in PCT Patent Application Publication No. WO/2020/247907, the entirety of which is incorporated by reference herein.

[0150] Example methods of assembling the guard member **104** are described in PCT Patent Application Publication No. WO/2022/087336 and WO/2023/059513, each of which is incorporated by reference herein in its entirety.

Exemplary Prosthetic Valves

[0151] FIGS. **6**A-**6**B show a prosthetic valve **200**, according to one example. The prosthetic valve **200** can be adapted to be implanted, with or without a docking device, in a native valve annulus, such as the native mitral valve annulus, native aortic annulus, native pulmonary valve annulus, etc. The prosthetic valve **200** can include a frame **212**, a valvular structure **214**, and a valve cover **216** (the valve cover **216** is removed in FIG. **6**A to show the frame structure).

[0152] The valvular structure **214** can include three leaflets **240**, collectively forming a leaflet structure (although a greater or fewer number of leaflets can be used), which can be arranged to collapse in a tricuspid arrangement. The leaflets **240** are configured to permit the flow of blood from an inflow end **222** to an outflow end **224** of the prosthetic valve **200** and block the flow of blood from the outflow end **224** to the inflow end **222** of the prosthetic valve **200**. The leaflets **240** can be secured to one another at their adjacent sides to form commissures **226** of the leaflet structure. The lower edge of valvular structure **214** desirably has an undulating, curved scalloped shape. By forming the leaflets **240** with this scalloped geometry, stresses on the leaflets **240** can be reduced, which in turn can improve durability of the prosthetic valve **200**. Moreover, by virtue of the scalloped shape, folds and ripples at the belly of each leaflet **240** (the central region of each leaflet), which can cause early calcification in those areas, can be eliminated or at least minimized. The scalloped geometry can also reduce the amount of tissue material used to form leaflet structure, thereby allowing a smaller, more even crimped profile at the inflow end of the prosthetic valve **200**. The leaflets **240** can be formed of pericardial tissue (for example, bovine pericardial tissue), biocompatible synthetic materials, or various other suitable natural or synthetic materials as known in the art and described in U.S. Pat. No. 6,730,118, which is incorporated by reference herein.

[0153] The frame **212** can be formed with a plurality of circumferentially spaced slots, or commissure windows **220** (three in the illustrated example) that are adapted to mount the commissures **226** of the valvular structure **214** to the frame. The frame **212** can be made of any of various suitable plastically expandable materials (for example, stainless steel, etc.) or self-expanding materials (for example, Nitinol) as known in the art. When constructed of a plastically expandable material, the frame **212** (and thus the prosthetic valve **200**) can be crimped to a radially compressed state on a delivery apparatus and then expanded inside a patient by an inflatable balloon or equivalent expansion mechanism. When constructed of a self-expandable material, the frame **212** (and thus the prosthetic valve **200**) can be crimped to a radially compressed state and restrained in the compressed state by insertion into a valve sheath or equivalent mechanism of a delivery apparatus. Once inside the body, the prosthetic valve **200** can be advanced from the delivery sheath, which allows the prosthetic valve **200** to expand to its functional size.
[0154] Suitable plastically expandable materials that can be used to form the frame **212** include, without limitation, stainless steel, a nickel-based alloy (for example, a cobalt-chromium or a nickel-cobalt-chromium alloy), polymers, or combinations thereof. In particularly examples, frame **212** can be made of a nickel-cobalt-chromium-molybdenum alloy, such as MP35N™ (tradename of SPS Technologies), which is equivalent to UNS R30035 (covered by ASTM F562-02). MP35N™/UNS R30035 comprises 35% nickel, 35% cobalt, 20% chromium, and 10% molybdenum, by weight. It has been found that the use of MP35N to form the frame **212** can provide superior structural results over stainless steel. In particular, when MP35N is used as the frame material, less material is needed to achieve the same or better performance in radial and crush force resistance, fatigue resistances, and corrosion resistance. Moreover, since less material is required, the crimped profile of the frame can be reduced, thereby providing a lower profile valve assembly for percutaneous delivery to the treatment location in the body.
[0155] As shown in FIG. **6**B, the valve cover **216** can include an outer portion **218** which can cover an entire outer surface of the frame **212**. In certain examples, as shown in FIG. **7**A, the valve cover **216** can also include an inner portion **228**. The inner portion **228** can cover an entire inner surface of the frame **212**, or alternatively, cover only selected portions of the inner surface of the frame **212**. In the depicted example, the inner portion **228** is formed by folding the valve cover **216** over the outflow end **224** of the frame **212**. In certain examples, a protective cover **236** comprising a high abrasion resistant material (for example, ePTFE, etc.) can be placed over the fold of the valve cover **216** at the outflow end **224**. In certain examples, similar protective cover **236** can be placed over the inflow end **222** of the frame. The valve cover **216** and the protective cover **236** can be affixed to the frame **212** by a variety of means, such as via sutures **230**.
[0156] As described herein, the valve cover **216** can be configured to prevent paravalvular leakage between the prosthetic valve **200** and the native valve, to protect the native anatomy, to promote tissue ingrowth, among some other purposes. For mitral valve replacement, due to the general D-shape of the mitral valve and relatively large annulus compared to the aortic valve, the valve cover **216** can act as a seal around the prosthetic valve **200** (for example, when the prosthetic valve **200** is sized to be smaller than the annulus) and allows for smooth coaptation of the native leaflets against the prosthetic valve **200**.
[0157] In various examples, the valve cover **216** can include a material that can be crimped for transcatheter delivery of the prosthetic valve **200** and is expandable to prevent paravalvular leakage around the prosthetic valve **200**. Examples of possible materials include foam, cloth, fabric, one or more synthetic polymers (for example, polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), expanded polytetrafluoroethylene (ePTFE), etc.), organic tissues (for example, bovine pericardium, porcine pericardium, equine pericardium, etc.), and/or an encapsulated material (for example, an encapsulated hydrogel).
[0158] In certain examples, the valve cover **216** can be made of a woven cloth or fabric possessing a plurality of floated yarn sections **232** (for example, protruding or puffing sections, also referred to

as “floats” hereinafter). Details of exemplary covered valves with a plurality of floats **232** are further described in U.S. Patent Publication Nos. US2019/0374337, US2019/0192296, and US2019/0046314, the disclosures of which are incorporated herein in their entireties for all purposes. In certain examples, the floated yarn sections **232** are separated by one or more horizontal bands **234**. In some examples, the horizontal bands **234** can be constructed via a leno weave, which can improve the strength of the woven structure. In some examples of the woven cloth, vertical fibers (for example, running along the longitudinal axis of the prosthetic valve **200**) can include a yarn or other fiber possessing a high level of expansion, such as a texturized weft yarn, while horizontal fibers (for example, running circumferentially around the prosthetic valve **200**) in a leno weave can include a low expansion yarn or fiber.

[0159] In some examples, the valve cover **216** can include a woven cloth resembling a greige fabric when assembled and under tension (for example, when stretched longitudinally on a compressed valve prior to delivery of a prosthetic valve **200**). When the prosthetic valve **200** is deployed and expanded, tension on floats **232** is relaxed allowing expansion of the floats **232**. In some examples, the valve cover **216** can be heat set to allow floats **232** to return to an enlarged, or puffed, space-filling form. In some examples, the number and sizes of floats **232** can be optimized to provide a level of expansion to prevent paravalvular leakage across the mitral plane (for example, to have a higher level of expansion thickness) and/or a lower crimp profile (for example, for delivery of the prosthetic valve). Additionally, the horizontal bands **234** can be optimized to allow for attachment of the valve cover **216** to the frame **212** based on the specific size or position of struts or other structural elements on the prosthetic valve **200**.

[0160] Further details of the prosthetic valve **200** and its components are described, for example, in U.S. Pat. Nos. 9,393,110 and 9,339,384, which are incorporated by reference herein. Additional examples of the valve cover are described in PCT Patent Application Publication No. WO/2020/247907.

[0161] As described above and illustrated in FIGS. **7**A-**7**B, the prosthetic valve **200** can be radially expanded and securely anchored within the docking device **100**.

[0162] In certain examples, the coil **102** of the docking device **100** in the deployed configuration can be movable between a first radially expanded configuration before the prosthetic valve **200** is radially expanded within the coil **102** and a second radially expanded configuration after the prosthetic valve **200** is radially expanded within the coil **102**. In the example depicted in FIGS. **7**A-**7**B, the coil **102** is in the second radially expanded configuration since the prosthetic valve **200** is shown in the radially expanded state.

[0163] As described herein, at least a portion of the coil **102**, such as the central region **108**, can have a larger diameter in the second radially expanded configuration than in the first radially expanded configuration (that is, the central region **108** can be further radially expanded by radially expanding the prosthetic valve **200**). As the central region **108** increases in diameter when the coil **102** moves from the first radially expanded configuration to the second radially expanded configuration, the functional turns in the central region **108** and the leading turn **106** can rotate circumferentially (for example, in clockwise or counter-clockwise direction when viewed from the stabilization turn **110**). Circumferential rotation of the functional turns in the central region **108** and the leading turn **106**, which can also be referred to as “clocking,” can slightly unwind the helical coil in the central region **108**. Generally, the unwinding can be less a turn, or less than a half turn (that is, 180 degrees). For example, the unwinding can be about 60 degrees and may be up to 90 degrees in certain circumstances. As a result, a distance between the proximal end **102***p* and the distal end **102***d* of the coil **102** measured along the central longitudinal axis of the coil **102** can be foreshorten.

[0164] In the example depicted in FIGS. **7**A-**7**B, the proximal end **105** of the guard member **104** is shown to be positioned distal to the proximal seating marker **121***p*. In other examples, after the prosthetic valve **200** is radially expanded within the coil **102**, the proximal end **105** of the guard member **104** can be positioned proximal to the proximal seating marker **121***p* (that is, the proximal seating marker **121***p* is covered by the guard member **104**) but remains distal to the ascending portion **110***b*.

Exemplary Retention Members

[0165] As described above, the retention member (for example, **114**) of the docking device can have a number of desired functions. For example, the textured outer surface of the retention member can be configured to encourage or promote tissue ingrowth. The tissue ingrowth can help reducing or preventing migration of the prosthetic valve and docking device relative to the native tissue. The retention member extending out of a proximal end of the guard member can resist motion of the guard member relative to the coil. Additionally, the retention member extending out of a distal end of the guard member can frictionally engage the prosthetic valve that is radially expanded within the docking device so as to anchor the prosthetic valve more securely. However, in certain circumstances, hemolysis may occur when some red blood cells shear against the rough textured outer surface of the retention braid, especially when there is a paravalvular leakage jet at the mitral valve. As described herein, the retention braid can be configured to mitigate or reduce the likelihood of hemolysis while still maintaining the desired functions noted above.

[0166] In one example, FIG. **8** depicts a section of a docking device **300** in a substantially straight delivery configuration, and FIGS. **8**A-**8**B show two cross-sections of the docking device **300**, respectively. As shown, the docking device **300** includes a coil **302** (similar to **102**), a tubular member **312** (similar to **112**) covering and fixedly attached to the coil **302**, and a retention member **314** covering at least a portion of the tubular member **312** (thus also covering a corresponding portion of the underlying coil **302**). As described above, the tubular member **312** can include ePTFE, which provides a biocompatible and relatively smooth surface. In some examples, the retention member **314** has a braided structure, for example, by interweaving a plurality of fibers in a diagonally overlapping pattern. In some examples, the retention member **314** has a woven structure, for example, by interlacing two sets of fibers, the warp and the weft, which are at about the right angle to each other. For clarity, the guard member (similar to **104**) of the docking device **300** is removed.

[0167] The retention member **314** can have a first axial segment **314***d* and a second axial segment **314***p* that are connected at a boundary **316**. As shown in FIG. **8**A, the second axial segment **314***p* encloses the full circumference of the coil **302** (and the tubular member **312**). Thus, the second axial segment **314***p* has a generally cylindrical shape. In contrast, as shown in FIG. **8**B, the first axial segment **314***d* encloses only a partial circumference of the coil **302** (and the tubular member **312**). Thus, the first axial segment **314***d* has a longitudinally truncated cylindrical shape (for example, a half shell shape). As a result, a circumferential portion of the coil **302** and the corresponding circumferential portion of the tubular member **312** (extending along the first axial segment **314***d*) are not covered by the retention member **314**.

[0168] In some examples, the partial circumference of the coil **302** enclosed by the first axial segment **314***d* has an arc angle (A1) between 90 degrees and 300 degrees, inclusive. In some examples, A1 is between 120 degrees and 270 degrees, inclusive. In one specific example, A1 is about 180 degrees.

[0169] Although not shown, the guard member of the docking device **300** can be attached to the tubular member **312** (and the underlying coil **302**) so that the guard member surrounds at least a mid-portion of the retention member **314**, similar to the examples depicted in FIGS. **5**A-**5**B and described above.

[0170] In certain examples, the second axial segment **314***p* is located proximal to the first axial segment **314***d*. For example, when the guard member of the docking device **300** is in the radially expanded state, the first axial segment **314***d* can extend out of a distal end of the guard member, and the second axial segment **314***p* can extend out of a proximal end of the guard member. In some examples, a proximal end of the second axial segment **314***p* can be situated at an ascending portion (similar to **110**_b_) of the coil **302**.

[0171] In certain examples, the first axial segment **314**_d_ is configured to have a predetermined angular position relative to the coil **302** and the tubular member **312**. Specifically, when the docking device **300** is deployed at the native valve, at least a portion of the first axial segment **314**_d_ is oriented to face radially inwardly so as to frictionally engage the prosthetic valve that is radially expanded within the docking device **300**, and a portion of the coil and the corresponding tubular member extending along and not covered by the first axial segment **314**_d_ are oriented to face radially outwardly. Because the uncovered portion of the tubular member generally has a smoother surface than the retention member, it will be less likely to damage the red blood cells and thus the risk of hemolysis can be reduced.

[0172] The retention member **314** can be prepared in a number of ways. For example, the initial retention member can have a tubular configuration (that is, having a cylindrical shape throughout its longitudinal length). A circumferential portion of the retention member can be removed by first cutting along two longitudinally parallel lines that are circumferentially spaced apart. The two parallel cut lines can extend from a distal end of the tubular retention member to the boundary **316**. Then, another circumferential cut across the two parallel cut lines can be made at the boundary **316** to remove the circumferential portion between the cut lines. The removal of the circumferential portion of the retention member can therefore create the longitudinally truncated cylindrical shape of the first axial segment **314**_d_. For attachment, the retention member **314** can slide over the tubular member **312**, and then be affixed to the tubular member **312** (and thus the underlying coil **302**) via sutures, as described below.

[0173] As another example, the initial retention member can similarly have a tubular configuration. A longitudinal cut of the retention member can be made to create a slit extending from a distal end of the tubular retention member to the boundary **316**. Then a circumferential cut connected to the slit and with a predefined arc angle can be made at the boundary **316**. The cut portion of the retention member can then be folded longitudinally in half. Such folding can therefore create the longitudinally truncated cylindrical shape of the first axial segment **314**_d_. For attachment, the retention member **314** can slide over the tubular member **312**, and then be affixed to the tubular member **312** (and the underlying coil **302**) via sutures, as described below.

[0174] The cutting in any of the examples described above can be laser cutting or other cutting means that sufficiently heat or fuse the fibers of the retention member **314** along the cutting edges so as to create seals to prevent fraying of the fibers.

[0175] As another example, the initial retention member can have a flat configuration. For example, both the first axial segment **314**_d_ and second axial segment **314**_p_ can be configured as rectangular sheets. The width of the second axial segment can be equal to the circumference of the tubular member **312**, and the width of the first axial segment is smaller than the circumference of the tubular member **312**. For attachment, the sheet of retention member **314** can be wrapped around the tubular member **312** (and the underlying coil **302**) so as to create the cylindrical shape of the second axial segment **314**_p_ and the longitudinally truncated cylindrical shape of the first axial segment **314**_d_. The retention member **314** can then be affixed to tubular member **312** (and the underlying coil **302**) via sutures, as described below.

[0176] In certain examples, the first axial segment **314**_d_ can be affixed to the coil **302** and the tubular member **312** via a plurality of stitches **318** extending along two longitudinal edges **320** of the first axial segment **314**_d_. For instance, the plurality of stitches **318** can be formed by a continuous in-and-out stitches extending through a thickness of the first axial segment **314**_d_ and the underlying tubular member **312**. The plurality of stitches **318** can have a very small inter-stitch distance (d**1**) to prevent fraying of the fibers at the edges **320** and detachment of the retention member **314** from the tubular member **312**. In some examples, d**1** is between 0.5 mm and 1.5 mm, inclusive. In one specific example, d**1** is about 1 mm.

[0177] To further illustrate, FIG. **10** is a side view of a portion of the first axial segment **314**_d_ when the docking device **300** is in a substantially straight configuration. As shown, the first axial segment **314***d* has a circumferential gap **322** exposing the underlying tubular member **312**. The gap **322** is formed between two longitudinal edges **320** of the first axial segment **314***d*. In this example, A1 is greater than 180 degrees (thus, the gap **322** has an arc angle that is less than 180 degrees). As shown, the first axial segment **314***d* is attached to the underlying coil and tubular member by a plurality of stitches **318** extending along the two longitudinal edges **320**.

[0178] In some examples, the second axial segment **314***p* can be affixed to the coil **302** via a spiral suture **324** The spiral suture **324** can extend along the entire length of the second axial segment **314***p* while rotating around a longitudinal axis of the second axial segment **314***p*. For example, the spiral suture **324** can be formed by a continuous in-and-out stitches extending through a thickness of the second axial segment **314***p* and the underlying tubular member **312**. The spiral suture **324** has a pitch (d**2**), which is the axial distance for the spiral suture **324** to complete one rotation. In some examples, d**2** is between 2 mm and 8 mm, or between 3 mm and 7 mm, or between 4 mm and 6 mm, all inclusive. In one specific example, d**2** is about 5 mm.

[0179] In another example, FIG. **9** depicts a section of a docking device **400** in a substantially straight delivery configuration, and FIGS. **9**A-**9**B show two cross-sections of the docking device **400**, respectively. Similarly, the docking device **400** includes a coil **402** (similar to **102**), a tubular member **412** (similar to **112**) covering and fixedly attached to the coil **402**, and a retention member **414** covering at least a portion of the tubular member **412** (thus also covering a corresponding portion of the underlying coil **402**). The retention member **414** can comprise a braided material or a woven material. For clarity, the guard member (similar to **104**) of the docking device **400** is removed.

[0180] The retention member **414** can have a first axial segment **414***d* and a second axial segment **414***p* that are connected at a boundary **416**. As shown in FIGS. **8**A-**8**B, both the first axial segment **414***d* and the second axial segment **414***p* enclose the full circumference of the tubular member **412** (and the underlying coil **402**). Thus, the entire retention member **414** has a generally cylindrical shape. In contrast to the second axial segment **414***p* which has a substantially uniform and textured outer surface in full circumference (360 degrees), the first axial segment **414***d* has two circumferential portions that exhibit different physical properties. Specifically, the first axial segment **414***d* includes a first circumferential portion **418** which has a textured (and rough) outer surface similar to the second axial segment **414***p*, and a second circumferential portion **420** which has a non-textured (and smooth) outer surface. In other words, the second circumferential portion **420** has a smoother outer surface than the first circumferential portion **418** (and the second axial segment **414***p*).

[0181] As described herein, the surface roughness can be measured based on industry standards such as ISO 4287-1:1984, DIN 4762, 4768, etc. For example, the surface roughness can be calculated based on measuring the average or root-mean-square of microscopic peaks and valleys across the surface. In some cases, the surface roughness of an object may be also measured by comparing the surface of the object to a known sample.

[0182] In some examples, the first circumferential portion **418** has an arc angle (A2) between 90 degrees and 300 degrees, inclusive. In some examples, A2 is between 120 degrees and 270 degrees, inclusive. In one specific example, A2 is about 180 degrees.

[0183] Although not shown, the guard member of the docking device **400** can be attached to the tubular member **412** and the coil **402** so that the guard member surrounds at least a mid-portion of the retention member **414**, similar to the examples depicted in FIGS. **5**A-**5**B and described above.

[0184] In certain examples, the second axial segment **414***p* is located proximal to the first axial segment **414***d*. For example, when the guard member of the docking device **400** is in the radially expanded state, the first axial segment **414***d* can extend out of a distal end of the guard member, and the second axial segment **414***p* can extend out of a proximal end of the guard member. In some examples, a proximal end of the second axial segment **414***p* can be situated at an ascending portion (similar to **110**_b_) of the coil **402**.

[0185] In certain examples, the first axial segment **414**_d_ is configured to have a predetermined angular position relative to the coil **402** and the tubular member **412**. Specifically, when the docking device **400** is deployed at the native valve, at least a portion of the first circumferential portion **418** is oriented to face radially inwardly so as to frictionally engage the prosthetic valve that is radially expanded within the docking device **400**, and at least a portion of the second circumferential portion **420** is oriented to face radially outwardly. Because the second circumferential portion **420** has a smoother surface than other portions (for example, **418**, **414**_p_) of the retention element, it will be less likely to damage the red blood cells and thus the risk of hemolysis can be reduced.

[0186] To further illustrate, FIG. **11** is a side view of the docking device **400** in the helical deployed configuration (for example, a plurality of helical turns **408**_p_, **408**_m_, **408**_d_ form a central region **408** similar to **108** of FIG. **5**A). As shown, the second axial segment **414**_p_ extends out of the proximal end of the guard member **404** and the first axial segment **414**_d_ extends out of a distal end of the guard member **404**. The second axial segment **414**_p_ has a textured outer surface around the full circumference. The first axial segment **414**_d_ has the first circumferential portion **418** (with a textured outer surface) and the second circumferential portion **420** (with a smoother outer surface). In this example, A2 is greater than 180 degrees (thus, the smooth-surfaced second circumferential portion **420** has an arc angle that is less than 180 degrees). Thus, the second circumferential portion **420** faces radially outwardly and at least a portion of the first circumferential portion **418** (on opposite side of **420**) faces radially inwardly. Additionally, the first circumferential portion **418** also covers the top and bottom surfaces of the helical turns (for example, **408**_p_, **408**_m_, **408**_d_). The textured first circumferential portion **418** between the helical turns can provide additional frictional force therebetween to resist their rotational movement relative to one another, for example, when the prosthetic valve is radially expanded within the central region **408**. Such additional frictional force between the helical turns can further facilitate retention of the prosthetic valve within the docking device **400**.

[0187] For attachment, the retention member **414** can slide over the tubular member **412**, and then be affixed to the tubular member **412** (and the underlying coil **402**) via sutures **422**, as depicted in FIG. **9**. In some examples, the suture **422** can have a spiral pattern, extending along the entire length of the retention member **414** while rotating around a longitudinal axis of the retention member **414**. The spiral suture **422** has a pitch (d**3**), which can be between 2 mm and 8 mm, or between 3 mm and 7 mm, or between 4 mm and 6 mm, all inclusive. In one specific example, d**3** is about 5 mm. In some examples, the spiral suture **422** can have a smaller pitch in the first axial segment **414**_d_ than in the second axial segment **414**_p_.

[0188] In the second axial segment **414**_p_, the suture **422** can extend through a thickness of the second axial segment **414**_p_ and the underlying tubular member **412** (for example, via in-and-out stitches). Thus, the suture **422** is exposed outside the second axial segment **414**_p_, and such exposed suture **422** can contribute to local roughness of the outer surface.

[0189] In the first axial segment **414**_d_, the suture **422** still extends through the tubular member **412** so as to stitch together the first axial segment **414**_d_ and the tubular member **412**. However, the suture **422** can extend through a thickness of the first circumferential portion **418**, but not through a thickness of the second circumferential portion **420**. In other words, the suture **422** is exposed outside the first circumferential portion **418**, but is hidden underneath the second circumferential portion **420**. Thus, the exposed suture **422** outside the first circumferential portion **418** can contribute to the local roughness, whereas the second circumferential portion **420** can maintain a smooth outer surface.

[0190] The retention member **414** can be prepared in a number of ways. For example, the initial retention member can have a tubular configuration with a braided structure which provides a substantially uniform texture (and roughness) over its outer surface. The smooth-surfaced second circumferential portion **420** can be created by targeted heating to fuse the braided material located at the second circumferential portion **420**.

[0191] FIGS. **12**A-**12**D illustrate one example method of preparing the retention member **414**. Specifically, FIG. **12**A depicts an example holder device **430** which has a longitudinally truncated cylindrical shape with a top opening **432**. The holder device **430** can comprise a material that has a low thermal conductivity, such as wood, plastics, fiberglass, etc. The inner diameter of the holder device **430** can be configured to match the outer diameter of the retention member **414**. FIG. **12**B shows placing the retention member **414** in the holder device **430**. As described above, the retention member **414** can be attached to and enclose the tubular member **412** and the coil **402** of the docking device **400**. At this stage, the retention member **414** has a fully textured outer surface. For example, FIG. **12**B shows a circumferential portion **420***a* of the retention member **414** exposed through the top opening **432** of the holder device **430** has a textured outer surface.

[0192] FIG. **12**C shows pressing a soldering iron **434** on the exposed circumferential portion **420***a* of the retention member **414**. The soldering iron **434** can be heated to a predetermined temperature and pressed on a target location for a predetermined duration so that the fibers at the target location can be fused together. The soldering iron **434** can move longitudinally along the top opening **432** until the exposed circumferential portion **420***a* is all heat treated to form a smooth membrane or film **420***b*, as illustrated in FIG. **12**D. Thus, the holder device **430** in this example masks the unexposed circumferential portion of the retention member **414** (that is, the circumferential portion covered by the holder device) from the heat treatment, thereby maintaining its textured (rough) outer surface.

[0193] Although the soldering iron **434** is shown in this example as a heating source, it is to be understood that other heating sources (for example, using an ultrasonic welding machine, or other types of heating apparatuses) can be used to fuse the fibers and create the smooth surface portion of the retention member.

Sterilization

[0194] Any of the systems, devices, apparatuses, etc. herein can be sterilized (for example, with heat/thermal, pressure, steam, radiation, and/or chemicals, etc.) to ensure they are safe for use with patients, and any of the methods herein can include sterilization of the associated system, device, apparatus, etc. as one of the steps of the method. Examples of heat/thermal sterilization include steam sterilization and autoclaving. Examples of radiation for use in sterilization include, without limitation, gamma radiation, ultra-violet radiation, and electron beam. Examples of chemicals for use in sterilization include, without limitation, ethylene oxide, hydrogen peroxide, peracetic acid, formaldehyde, and glutaraldehyde. Sterilization with hydrogen peroxide may be accomplished using hydrogen peroxide plasma, for example.

Additional Examples of the Disclosed Technology

[0195] In view of the above-described implementations of the disclosed subject matter, this application discloses the additional examples enumerated below. It should be noted that one feature of an example in isolation or more than one feature of the example taken in combination and, optionally, in combination with one or more features of one or more further examples are further examples also falling within the disclosure of this application.

[0196] Example 1. A docking device for securing a prosthetic valve at a native valve, the docking device comprising: a coil comprising a plurality of helical turns when deployed at the native valve; and a retention member covering at least a portion of the coil, wherein the retention member comprises a first axial segment which encloses only a partial circumference of the coil.

[0197] Example 2. The docking device of any example herein, particularly example 1, wherein the retention member comprises a second axial segment that encloses a full circumference of the coil.

[0198] Example 3. The docking device of any example herein, particularly example 2, wherein the second axial segment is proximal to the first axial segment.

[0199] Example 4. The docking device of any example herein, particularly example 3, further comprising an expandable member surrounding at least a portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state, wherein when the expandable member is in the radially expanded state, the first axial segment of the retention member extends out of a distal end of the expandable member, and the second axial segment of the retention member extends out of a proximal end of the expandable member.

[0200] Example 5. The docking device of any example herein, particularly any one of examples 3-4, wherein a proximal end of the second axial segment is situated at an ascending portion of the coil, wherein when the coil is deployed at the native valve, the ascending portion forms an angle relative to a plane defined by the helical turns, wherein the angle is between 45 degrees and 90 degrees, inclusive.

[0201] Example 6. The docking device of any example herein, particularly any one of examples 1-5, wherein retention member comprises a braided material.

[0202] Example 7. The docking device of any example herein, particularly any one of examples 1-5, wherein the retention member comprises a woven material.

[0203] Example 8. The docking device of any example herein, particularly any one of examples 1-7, wherein the first axial segment is affixed to the coil via a plurality of stitches extending along two longitudinal edges of the first axial segment.

[0204] Example 9. The docking device of any example herein, particularly example 8, wherein the plurality of stitches has an inter-stitch distance between 0.5 mm and 1.5 mm, inclusive.

[0205] Example 10. The docking device of any example herein, particularly any one of examples 1-9, wherein the partial circumference of the coil enclosed by the first axial segment of the retention member has an arc angle between 90 degrees and 300 degrees, inclusive.

[0206] Example 11. The docking device of any example herein, particularly example 10, wherein the arc angle is between 120 degrees and 270 degrees, inclusive.

[0207] Example 12. The docking device of any example herein, particularly example 11, wherein the arc angle is 180 degrees.

[0208] Example 13. The docking device of any example herein, particularly any one of examples 1-12, wherein when the coil is deployed at the native valve, at least a portion of the first axial segment faces radially inwardly so as to frictionally engage the prosthetic valve that is radially expanded within a central region defined by the helical turns, wherein a portion of the coil extending along and not covered by the first axial segment faces radially outwardly.

[0209] Example 14. A docking device for securing a prosthetic valve at a native valve, the docking device comprising: a coil comprising a plurality of helical turns when deployed at the native valve; and a retention member covering at least a portion of the coil, wherein the retention member comprises a first axial segment, the first axial segment comprising a first circumferential portion and a second circumferential portion, wherein the second circumferential portion has a smoother outer surface than the first circumferential portion.

[0210] Example 15. The docking device of any example herein, particularly example 14, wherein the retention member comprises a second axial segment enclosing a full circumference of the coil, wherein the second axial segment has at least the same roughness as the first circumferential portion.

[0211] Example 16. The docking device of any example herein, particularly example 15, wherein the second axial segment is proximal to the first axial segment.

[0212] Example 17. The docking device of any example herein, particularly example 16, further comprising an expandable member surrounding at least a portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state, wherein when the expandable member is in the radially expanded state, the first axial segment of the retention member extends out of a distal end of the expandable member, and the second axial segment of the retention member extends out of a proximal end of the expandable

member.
[0213] Example 18. The docking device of any example herein, particularly any one of examples 16-17, wherein a proximal end of the second axial segment is situated at an ascending portion of the coil, wherein when the coil is deployed at the native valve, the ascending portion forms an angle relative to a plane defined by the helical turns, wherein the angle is between 45 degrees and 90 degrees, inclusive.
[0214] Example 19. The docking device of any example herein, particularly any one of examples 14-18, wherein retention member comprises a braided material.
[0215] Example 20. The docking device of any example herein, particularly any one of examples 14-18, wherein the retention member comprises a woven material.
[0216] Example 21. The docking device of any example herein, particularly any one of examples 14-20, wherein the first axial segment is affixed to the coil via a spiral suture extending along and around a longitudinal axis of the coil, wherein the spiral suture has a pitch between 4 mm and 6 mm, inclusive.
[0217] Example 22. The docking device of any example herein, particularly example 21, wherein the spiral suture extends through a thickness of the first circumferential portion, but does not extend through a thickness of the second circumferential portion.
[0218] Example 23. The docking device of any example herein, particularly any one of examples 14-22, wherein the first circumferential portion has an arc angle between 90 degrees and 300 degrees, inclusive.
[0219] Example 24. The docking device of any example herein, particularly example 23, wherein the arc angle is between 120 degrees and 270 degrees, inclusive.
[0220] Example 25. The docking device of any example herein, particularly example 24, wherein the arc angle is 180 degrees.
[0221] Example 26. The docking device of any example herein, particularly any one of examples 14-25, wherein when the coil is deployed at the native valve, at least a portion of the first circumferential portion faces radially inwardly so as to frictionally engage the prosthetic valve that is radially expanded within a central region defined by the helical turns, wherein at least a portion of the second circumferential portion faces radially outwardly.
[0222] Example 27. A method for assembling a docking device configured to receive a prosthetic valve, the method comprising: preparing a retention member having a first axial segment and a second axial segment; and attaching the retention member to a coil of the docking device so that the first axial segment encloses only a partial circumference of the coil and the second axial segment encloses a full circumference of the coil, wherein the coil is movable from a substantially straight configuration to a helical configuration.
[0223] Example 28. The method of any example herein, particularly example 27, wherein the act of preparing the retention member comprises receiving the retention member in a tubular configuration, and removing a circumferential portion of the first axial segment.
[0224] Example 29. The method of any example herein, particularly example 27, wherein the act of preparing the retention member comprises receiving the retention member in a tubular configuration, cutting the first axial segment longitudinally, and folding longitudinally the first axial segment in half.
[0225] Example 30. The method of any example herein, particularly example 27, wherein the act of preparing the retention member comprises receiving the retention member in a flat configuration, wherein the first axial segment has a smaller width than the second axial segment, wherein the act of attaching comprises wrapping the retention member around the coil.
[0226] Example 31. The method of any example herein, particularly any one of examples 27-30, where the act of attaching comprises stitching the retention member to the coil along two longitudinal edges of the first axial segment.
[0227] Example 32. The method of any example herein, particularly any one of examples 27-31,

wherein the act of attaching comprises orienting the first axial segment relative to the coil such that when the coil is moved to the helical configuration, at least a portion of the first axial segment faces radially inwardly, and a portion of the coil extending along and not covered by the first axial segment faces radially outwardly.

[0228] Example 33. The method of any example herein, particularly any one of examples 27-32, further comprising attaching an expandable member to the coil so that the expandable member surrounds at least a mid-portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state.

[0229] Example 34. A method for assembling a docking device configured to receive a prosthetic valve, the method comprising: preparing a retention member having a tubular configuration, the retention member having a first axial segment and a second axial segment; and attaching the retention member to a coil of the docking device so that the retention member encloses a full circumference of the coil, the coil being movable from a substantially straight configuration to a helical configuration, wherein the first axial segment comprises a first circumferential portion and a second circumferential portion, wherein the second circumferential portion has a smoother outer surface than the first circumferential portion.

[0230] Example 35. The method of any example herein, particularly example 34, wherein the retention member comprises a braided material, wherein the act of preparing the retention member comprises heating the second circumferential portion so as to fuse the braided material thereof.

[0231] Example 36. The method of any example herein, particularly example 35, wherein the act of heating comprises applying soldering or ultrasonic welding to the second circumferential portion.

[0232] Example 37. The method of any example herein, particularly any one of examples 35-36, wherein the act of preparing the retention member comprises masking the first circumferential portion from the heating.

[0233] Example 38. The method of any example herein, particularly any one of examples 34-37, wherein the act of attaching comprises affixing the first axial segment to the coil via a spiral suture extending along and around a longitudinal axis of the coil, wherein the spiral suture extends through a thickness of the first circumferential portion, but does not extend through a thickness of the second circumferential portion.

[0234] Example 39. The method of any example herein, particularly any one of examples 34-38, wherein the act of attaching comprises orienting the first axial segment relative to the coil such that when the coil is moved to the helical configuration, at least a portion of the first circumferential portion faces radially inwardly, and at least a portion of the second circumferential portion faces radially outwardly.

[0235] Example 40. The method of any example herein, particularly any one of examples 34-39, further comprising attaching an expandable member to the coil so that the expandable member surrounds at least a mid-portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state.

[0236] Example 41. An implant assembly comprising: a radially expandable and compressible prosthetic valve; and a docking device configured to receive the prosthetic valve, wherein the docking device is according to any one of examples 1-26.

[0237] Example 42. A method for implanting a prosthetic valve, the method comprising: deploying a docking device at a native valve; and deploying the prosthetic valve within the docking device, wherein the docking device is according to any one of examples 1-26.

[0238] Example 43. A method comprising sterilizing the docking device of any example herein, particularly any one of examples 1-26.

[0239] Example 44. A method of treating a heart on a simulation, the method comprising: deploying a docking device at a target location; and deploying a prosthetic valve within the docking device; wherein the docking device is according to any one of examples 1-26.

[0240] The features described herein with regard to any example can be combined with other

features described in any one or more of the other examples, unless otherwise stated. For example, any one or more of the features of one docking device can be combined with any one or more features of another docking device. As another example, any one or more features of one guard member can be combined with any one or more features of another guard member.
[0241] In view of the many possible examples to which the principles of the disclosed technology may be applied, it should be recognized that the illustrated examples are only preferred examples of the technology and should not be taken as limiting the scope of the disclosure. Rather, the scope of the claimed subject matter is defined by the following claims and their equivalents.

Claims

1. A docking device for securing a prosthetic valve at a native valve, the docking device comprising: a coil comprising a plurality of helical turns when deployed at the native valve; and a retention member covering at least a portion of the coil, wherein the retention member comprises a first axial segment which encloses only a partial circumference of the coil.
2. The docking device of claim 1, wherein the retention member comprises a second axial segment that encloses a full circumference of the coil.
3. The docking device of claim 2, wherein the second axial segment is proximal to the first axial segment.
4. The docking device of claim 3, further comprising an expandable member surrounding at least a portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state, wherein when the expandable member is in the radially expanded state, the first axial segment of the retention member extends out of a distal end of the expandable member, and the second axial segment of the retention member extends out of a proximal end of the expandable member.
5. The docking device of claim 1, wherein retention member comprises a braided material.
6. The docking device of claim 1, wherein the retention member comprises a woven material.
7. The docking device of claim 1, wherein the first axial segment is affixed to the coil via a plurality of stitches extending along two longitudinal edges of the first axial segment.
8. A docking device for securing a prosthetic valve at a native valve, the docking device comprising: a coil comprising a plurality of helical turns when deployed at the native valve; and a retention member covering at least a portion of the coil, wherein the retention member comprises a first axial segment, the first axial segment comprising a first circumferential portion and a second circumferential portion, wherein the second circumferential portion has a smoother outer surface than the first circumferential portion.
9. The docking device of claim 8, wherein the retention member comprises a second axial segment enclosing a full circumference of the coil, wherein the second axial segment has at least the same roughness as the first circumferential portion.
10. The docking device of claim 9, wherein the second axial segment is proximal to the first axial segment.
11. The docking device of claim 10, further comprising an expandable member surrounding at least a portion of the retention member, wherein the expandable member is movable between a radially compressed state and a radially expanded state, wherein when the expandable member is in the radially expanded state, the first axial segment of the retention member extends out of a distal end of the expandable member, and the second axial segment of the retention member extends out of a proximal end of the expandable member.
12. The docking device of claim 8, wherein the first axial segment is affixed to the coil via a spiral suture extending along and around a longitudinal axis of the coil.
13. The docking device of claim 12, wherein the spiral suture extends through a thickness of the first circumferential portion, but does not extend through a thickness of the second circumferential

portion.

14. A method for assembling a docking device configured to receive a prosthetic valve, the method comprising: preparing a retention member having a first axial segment and a second axial segment; and attaching the retention member to a coil of the docking device so that the first axial segment encloses only a partial circumference of the coil and the second axial segment encloses a full circumference of the coil, wherein the coil is movable from a substantially straight configuration to a helical configuration.

15. The method of claim 14, wherein the act of preparing the retention member comprises receiving the retention member in a tubular configuration, and removing a circumferential portion of the first axial segment.

16. The method of claim 14, wherein the act of preparing the retention member comprises receiving the retention member in a tubular configuration, cutting the first axial segment longitudinally, and folding longitudinally the first axial segment in half.

17. The method of claim 14, wherein the act of preparing the retention member comprises receiving the retention member in a flat configuration, wherein the first axial segment has a smaller width than the second axial segment, wherein the act of attaching comprises wrapping the retention member around the coil.

18. A method for assembling a docking device configured to receive a prosthetic valve, the method comprising: preparing a retention member having a tubular configuration, the retention member having a first axial segment and a second axial segment; and attaching the retention member to a coil of the docking device so that the retention member encloses a full circumference of the coil, the coil being movable from a substantially straight configuration to a helical configuration, wherein the first axial segment comprises a first circumferential portion and a second circumferential portion, wherein the second circumferential portion has a smoother outer surface than the first circumferential portion.

19. The method of claim 18, wherein the retention member comprises a braided material, wherein the act of preparing the retention member comprises heating the second circumferential portion so as to fuse the braided material thereof.

20. The method of claim 18, wherein the act of attaching comprises affixing the first axial segment to the coil via a spiral suture extending along and around a longitudinal axis of the coil, wherein the spiral suture extends through a thickness of the first circumferential portion, but does not extend through a thickness of the second circumferential portion.